Proposal Design Report

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Evidence-Traceable Surface Quality Monitoring for Recycled PLA Filament

Document field	Status
Client context	BioLoop Manufacturing Pty Ltd
Discipline	ENG5105 Electrical Engineering
Deliverable	Proposal design report
Selected quality parameter	Visible rPLA filament surface quality
Selected concept	Plan A: ALC-IMX258-LY0001-ZB V1 USB 2.0 camera module, Raspberry Pi 4 vision, size and specimen-relative position, and STM32 supervision and safe I/O. Interim implementation: ESP32S3 OV3660 acquisition, same downstream chain
Concept status	Plan A remains the selected concept, using the ALC-IMX258-LY0001-ZB V1 module. Every measurement in this report comes from the Plan B interim path; no Plan A measurement exists yet
Evidence cut-off	27 August 2026
Evidence status	Proposal with bounded integration evidence; real-material calibration and defect-performance validation remain pending

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Executive Summary

BioLoop requires a laboratory-scale system for monitoring visible surface defects in recycled polylactic acid (rPLA) filament, including bubbles, roughness, contamination, cracks and inclusions (Monash University, 2026d). This report selects Plan A: an ALC-IMX258-LY0001-ZB V1 USB 2.0 camera imaging a stationary, indexed filament sample, with Raspberry Pi 4 vision and specimen-relative position estimation, plus STM32 control of lighting, watchdog and inhibited downstream output. The supplier-listed sensor, video modes, focus options and fields of view remain unverified on the Raspberry Pi, and the rolling shutter is suitable only while the fixture is stationary (Supplier product parameter sheet, 2026). Until Plan A hardware is available, university-supplied ESP32S3/OV3660 hardware operates as the Plan B acquisition front end while preserving the downstream Pi, STM32 and evidence architecture (Monash University, 2026c). Current evidence is limited to Windows-host integration without rPLA in the region of interest; it demonstrates frame continuity, not Raspberry Pi deployment, physical STM32 I/O, defect accuracy, minimum detectable size or industrial readiness. All numerical requirements therefore remain proposed. Final acceptance requires controlled optics, optical and coordinate calibration, labelled real-material specimens, independent validation and documented requirement review; location will be reported only as an approximate fixture-referenced position within the calibrated fixed view.

Acknowledgement of Country

Monash University recognises that its Australian campuses are located on the unceded lands of the people of the Kulin Nations, and pays its respects to their Elders, past and present.

1. Background and Purpose

1.1 Client and manufacturing context

BioLoop's proposed manufacturing hub is meant to produce rPLA and rPLA-composite filament for Australian education, engineering, manufacturing and consumer markets. Product monitoring sits inside a wider circular-manufacturing system, alongside automation, resource efficiency, safety and economic viability (Monash University, 2026d). Filament only reaches a customer after extrusion, cooling, handling and spooling, so a visible anomaly caught late can mean rework, a rejected product, or a customer who no longer trusts the batch. Watching the line as it runs moves that decision point earlier.

Recycled PLA is not one uniform input. Published comparisons report that feedstock origin, recycling history and processing conditions can all affect the properties of recycled filament and printed products (Hidalgo-Carvajal et al., 2023; Wei & Bähr, 2024). Moisture can influence morphology and tensile behaviour in PLA-containing fused-filament material as well (Quader et al., 2024). Those findings justify monitoring. They do not let a camera diagnose moisture, degradation or extrusion settings from appearance alone. A pit or a colour patch is evidence of a visible condition, and nothing more; its process cause stays a hypothesis until it is checked against independent process and material data.

The circular-economy argument needs the same restraint. Life-cycle studies of distributed filament production and closed-loop PLA recycling show why recovered material deserves serious consideration, and they also show that the benefits depend on system boundaries, energy and process choices (Caceres-Mendoza et al., 2023; Zhao et al., 2018). This proposal claims no completed life-cycle assessment. Its purpose is narrower: get surface-quality evidence in front of a person, a quality system or a later plant controller early enough to act on.

1.2 Purpose, question and boundary

The report asks:

> Can a low-cost machine-vision system detect and grade visible anomalies on rPLA filament under representative laboratory conditions, while withholding unsafe or uncalibrated outputs?

The system has to make useful detections, and their validity has to be established independently. A stream that looks live can still be stale, badly lit or geometrically uncalibrated. That is why acquisition health and evidence status are design functions here, not interface decoration.

Table 1 fixes the investigation boundary before any concepts get compared.

Table 1. Proposed project scope.

In scope	Outside the present claim boundary
Visible bubbles or pits, cracks or scratches, contamination or inclusions, and abnormal surface texture identified from controlled images	Engineering roughness parameters such as Ra without a validated profilometric or structured-light reference
Laboratory-scale real-time acquisition, candidate detection, event recording, alarm presentation and safe-output requests on a held filament sample	Internal void metrology, polymer chemistry identification, or proof of a defect's process cause
Optical calibration followed by labelled real-rPLA testing	Uncalibrated millimetre dimensions, and any longitudinal position along a running line
Basic fault detection, and the interface through which a larger automated process would integrate it, as required by the brief (Monash University, 2026d)	Safety certification, autonomous production release, or industrial deployment claims

So the system produces a traceable candidate record and a decision aid for the operator. Automatic rejection is not authorised yet.

The optics settle it for this design. A 1.75 mm strand and a 0.20 mm target feature need magnification, and magnification costs depth of field: the working geometry in Section 5.2 leaves roughly 1.4 mm of focus depth, which a fixture can hold and a moving strand cannot. Real-time acquisition is unaffected — frames stream, the health gate runs and a preliminary result appears within a second for each illumination branch — but the sample is stationary while the shutter is open. The design basis is therefore a held, indexed filament sample, and the requirement set follows from it. R4 is a capture-to-result target rather than a line-speed one, and no requirement asks where along a strand a defect sits.

That choice does narrow the brief, and claiming otherwise would be the easier position rather than the true one. Three of the brief's demands survive intact: the system acquires and decides in real time, it detects, grades and requests rejection, and it is not asked for a longitudinal location, which the brief never requests (Monash University, 2026d). The fourth does not survive. Under the same team reading recorded above, a system suitable for integration into an industrial filament manufacturing process has to inspect material that is running, and this one inspects material that is held. What this project delivers is therefore a bench-scale instrument that establishes the measurement — optics, calibration, classification, evidence chain and safe-output path — on specimens drawn from the process rather than on the process itself.

Naming that gap is what makes it addressable, and it is an optical problem before it is a software one. Freezing motion at this magnification needs either a much shorter exposure with correspondingly more light, or a sensor and lens geometry that does not pay for scale in depth of field, because a running strand needs more focus depth than a fixture does. Everything above the optical layer is written to survive that change: the requirement set, the source-independent contracts, the supervision design, the calibration protocol and the validation method all hold when the head changes. Continuous in-line inspection therefore sits on the consultation list in Section 3 as a scoped extension, and it needs a different optical head rather than a different report.

2. Literature Review and Design Implications

2.1 What surface images can establish

Previous research indicates that the source of recycled material, repeated processing and moisture conditions can affect the properties and surface appearance of recycled PLA filament (Hidalgo-Carvajal et al., 2023; Quader et al., 2024; Wei & Bähr, 2024). These findings support the need for surface monitoring but do not provide direct visual ground truth for this project. A visible pit, crack, inclusion or color variation may indicate an abnormal surface condition, but its underlying cause cannot be determined for an image alone without additional material or process evidence.

Research on industrial anomaly detection also shows that model performance depends strongly on the image-acquisition environment and the representativeness of the dataset (Carvalho et al., 2024; Heckler-Kram et al., 2026; Wang et al., 2024).Results obtained from other materials or controlled datasets cannot be directly categorized and challenging optical conditions, while the final defect definitions must be established through labelled rPLA specimens collected under the project's own imaging configuration.

2.2 Optics, illumination and acquisition

The cylindrical and often reflective surface of rPLA filament creates several imaging challenges. Specular highlights can move as the filament rotates, while a single camera view cannot provide complete circumferential coverage. Changes in illumination angle, working distance of exposure may also alter the appearance of the same surface feature and introduce misleading image artefacts.Therefore, controlled lighting and repeatable optical geometry are essential for reliable surface inspection (Krassnig et al., 2025; Verspeek et al., 2024).

The proposed optical setup combines diffuse bright-field illumination with low-angle dark-field illumination. Diffuse bright-field lighting is intended to improve colour and texture contrast while low-angle dark-field lighting is intended to highlight surface relief, scratches and small irregularities through shadows. A diffuser, matter background and external enclosure will be used to reduce glare and minimise interference from ambient light. The region of interest, working distance ,focus and exposure will be fixed after calibration so that images collected under different conditions remain comparable.

Plan A uses a configurable close-range USB camera. The supplier lists several resolutions, focus options and nominal fields of view, but these specifications must be verified after the camera is installed in the final optical enclosure (Supplier product parameter sheet, 2026). Plan B uses the university-supplied OV3660 camera as an interim acquisition device (OmniVision Technologies, 2011; Espressif Systems, 2026; Seeed Studio, 2026). Because Plan B provides less control over the field of view, compression and optical settings, its working distance, illumination uniformity, glare response and achievable image quality will be treated as experimental conditions. No defect-

detection performance will be claimed until the lighting configuration and optical geometry have been calibrated and tested using representative rPLA samples.

2.3 Alternatives to area-scan surface imaging

Three optical measurement approaches were considered because they provide different types of quality information. Area- or line-scan imaging can retain spatial information about visible features such as colour variations, cracks, inclusions and abnormal surface texture, although its performance is sensitive to illumination and viewing geometry (Krassnig et al., 2025). Multiaxis optical diameter sensing can measure filament diameter and shape along a spool but it does not independently identify visible surface defects (Petsiuk & Pearce, 2021). Laser-line profilometry can provide quantitative surface height and topographical information, but it requires additional optical hardware, calibration and processing resources (Lafirenza et al., 2023).

Considering the project brief, available hardware and semester timeframe, area-scan imaging was selected as the primary method for detecting visible surface anomalies. The auxiliary photodiode channel will only be used to support material-presence and optical-health monitoring; it will not be treated as a surface-defect classifier. Diameter sensing and laser profilometry remain possible future extensions if stakeholder consultation identifies dimensional variation or engineering surface roughness as primary requirements,

2.4 Processing roles and evidence flow

Separating acquisition, processing and supervision creates clearer interfaces and prevents one device from becoming responsible for every system function. Edge-computing research supports performing time-sensitive image processing near the monitored equipment while maintaining clear boundaries between acquisition, analysis and control functions (Katsigiannis & Mykoniatis, 2024).

In the interim configuration, the ESP32S3 and OV3660 cameras acquire and transmit images. The Raspberry Pi 4 is responsible for OpenCV processing, event recording, data storage and the graphical user interface, while the STM32 provides deterministic capture gating, lighting control watchdog monitoring and safe-output supervision.The available Raspberry Pi and STM32 documentation indicates that these platforms provide the necessary processing and interface resource (Raspberry Pi Ltd, 2026a; STMicroelectronics, 2026). However, component capability does not demonstrate completed system performance.

This separation also establishes a clear response to system faults. If image acquisition fails or a frame becomes stale,the processing service should enter a fault or reconnecting state, while the STM32 keeps the physical reject or stop output inhibited. Timestamps, frame-age information, calibration identifiers and system-health states must therefore remain attached to each recorded event so that later results can be traced to the conditions under which they were produced

2.5 Validation implications

Adjacent frames captured from the same specimen or defect event are highly correlated. If near-duplicate frames are divided between the development and test datasets, the reported performance may become unrealistic and optimistic. Previous research identifies this form of data leakage as a significant threat to reproducible machine-learning evaluation particularly for video and sequential image data (Figueiredo & Mendes, 2024; Kapoor & Narayanan, 2023).

The validation dataset will therefore be separated by physical spool, specimen, batch or complete defect event rather than by individual frame. All images from the same event must remain entirely within either the development colour, surface gloss, illumination mode, light level, viewing angle and sample positioning.

Performance will be evaluated at the defect-event level using precision, recall and F1 score, together with false alarms per specimen, missed events, processing latency and stability across the declared illumination modes and viewing angle,. Results will be reported separately for different optical conditions instead of being reduced to one combined accuracy value. Sample counts and uncertainty will be reported alongside these results, and no performance value will be claimed until testing has been complete

Adjacent video frames are highly correlated. If near-duplicate frames from one defect land in both training and test data, reported performance comes out optimistic; broader machine-learning work identifies leakage as a reproducibility threat (Kapoor & Narayanan, 2023), and video-object-detection research shows the specific risk that spatiotemporal correlation creates (Figueiredo & Mendes, 2024). Validation therefore separates data by physical spool or batch and by defect event, never by shuffling individual frames at random.

Accuracy on its own would hide what rare defects cost. The proposed assessment reports event-level precision, recall and F1 score, false alarms per specimen, missed events, latency, and stability across the declared lighting modes and viewing angles. Sample counts and uncertainty go with those values. None of them exists yet.

Table 2 turns these literature findings into design decisions and names the project-specific evidence still missing.

Table 2. Literature-to-design transfer.

Evidence cluster	Design implication	Evidence still required for this project
rPLA variability and moisture (Hidalgo-Carvajal et al., 2023; Quader et al., 2024; Wei & Bähr, 2024)	Treat visible anomalies as observations, not automatic root-cause diagnoses	Labelled real-rPLA specimens linked to documented conditions
Industrial anomaly and multi-view datasets (Carvalho et al., 2024; Heckler-Kram et al., 2026; Wang et al., 2024)	Freeze optics and record viewpoint, lighting and dataset provenance	Representative images from the final fixture
Dynamic imaging and surface inspection (Krassnig et al., 2025; Sundarrajan & Rajendran, 2023; Verspeek et al., 2024)	Control exposure, motion, geometry and illumination; test coverage limits	Speed challenge and circumferential-coverage study
Optical diameter sensing and laser profilometry (Lafirenza et al., 2023; Petsiuk & Pearce, 2021)	Retain them as requirement-driven alternatives, not capabilities implied by area-scan images	Stakeholder priority, comparative test plan and approved cost/schedule change before any scope extension
Leakage and correlated video (Figueiredo & Mendes, 2024; Kapoor & Narayanan, 2023)	Split validation by independent spool/batch/event	Pre-registered dataset split and held-out test results

3. Stakeholders, Needs and Proposed Requirements

3.1 Stakeholder evidence status

Nobody has handed us a record of talking to a stakeholder, so this report has none. Everything below comes from the BioLoop brief, the discipline instructions and the literature. None of it should be read as something a client said, a participant agreed to, or a meeting settled. The distinction matters because the rubric asks for engagement and for requirements that actually shape the design (Monash University, 2026a, 2026b). We can write down what ought to be discussed. We cannot invent what someone said.

Table 3. Stakeholder map and provisional design response.

Stakeholder group	Need inferred from brief or system use	Proposed design response	Current status
BioLoop/client representative	Traceable quality evidence, cost awareness and a path to automated integration	Requirement register, event records, API and explicit acceptance gates	Brief-derived; confirmation pending
Production operator	Clear health state, actionable alarms and no silent stale-data decision	Single-screen status, candidate overlay, alarm history and inhibited output when unhealthy	Proposed
Quality assurance personnel	Reviewable images, batch context, repeatable classification and export	Versioned calibration/model identifiers, immutable event fields and dataset export	Proposed
Maintenance and electrical personnel	Diagnosable faults, serviceable interfaces and safe recovery	Camera/processor/lighting health, watchdog states and manual reset rules	Proposed
Laboratory supervisor and technical staff	Safe, testable use of supplied equipment	Enclosed optical path, low-voltage prototype domains and staged fault tests	Brief- and equipment-derived
Filament users	Consistent product information without unsupported quality claims	Evidence-backed grading only after validation; retain uncertain events for review	Literature-derived; confirmation pending

What we can deliver at this stage is the engagement plan itself. M1 owns the plan.

3.2 Proposed requirements and verification

Every numeric threshold in Table 4 carries the `PROPOSED_REQUIREMENT` label.

Table 4. Proposed requirements register.

ID	Proposed requirement	Stakeholder/source basis	Planned verification	Status or present gap
R1	Detect at least three agreed classes of visible surface anomaly	Defect families from the project brief (Monash University, 2026d); class count is team-proposed and awaits stakeholder confirmation	Held-out, event-level labelled test	`PENDING_REAL_MATERIAL`
R2	For each accepted R1 class, target the declared class-specific characteristic dimension at or above 0.20 mm	Team engineering target under the discipline requirement for measurable specifications (Monash University, 2026a); stakeholder confirmation pending	Calibrated target and labelled specimens using the definitions below	`PENDING_REAL_MATERIAL`; optical scale unavailable
R3	For every accepted R1 class whose events meet R2, achieve event-level precision and recall of at least 90% per class	Team performance target under (Monash University, 2026a); false-alarm and miss tolerance await stakeholder confirmation	Independent spool/batch/event split; report per-class counts and confidence intervals	`PENDING_REAL_MATERIAL`; no accuracy result exists
R4	Return a preliminary result within 1 s, measured per illumination branch from the start of its soft-start	Team responsiveness target under (Monash University, 2026a); operator expectation awaits stakeholder confirmation	Timed capture-to-display runs on the final fixture	Proposed; the interim continuous path sustains 10.77 frames/s at SVGA, which bounds preview responsiveness but is not this measure
R5	At P95, assert the alarm within 500 ms of the first frame in which a marked defect is clearly visible with illumination health already confirmed	Team alarm target under (Monash University, 2026a); response expectation awaits stakeholder confirmation	Synchronously film the scene reference and GUI/output; include confirmation, processing, communications and indication	Proposed; current brightness-control path timing is not this measure
R6	Store defect type, size, confidence,	Data and visualisation need from the brief (Monash University,	Schema, file and record audit	Partly implemented; live evidence images and complete optical provenance

ID	Proposed requirement	Stakeholder/source basis	Planned verification	Status or present gap
	evidence image, sample and view identifiers, time and batch identifier with the required calibration provenance	2026d), made testable under (Monash University, 2026a)		remain pending, and size stays null until calibration
R7	During a 30 min run, keep confirmed-event record loss below 1%, where `loss = (N_expected - N_complete) / N_expected` and a complete record has its durable database row plus required evidence image	Team reliability target under (Monash University, 2026a); stakeholder tolerance pending	Reconcile frozen expected event IDs against durable database rows, image files and export; report frame drops separately	Proposed; drop instrumentation is now in the service, but the final hardware test remains pending
R8	Enter a safe hold or fault state after camera, lighting or processor failure	Fault-detection/integration need from the brief (Monash University, 2026d); team safety interpretation, with later risk assessment required	Fault injection and output-state observation on final wiring	Software/network path evidence only; physical safe I/O pending
R9	Store and serve only images confined to the enclosed filament tunnel, with local-only storage and external publication disabled by default	Team privacy-by-design control; legal and stakeholder retention/access rules remain pending	Audit every final-evidence image and the storage/access configuration; accept zero outside-tunnel or person-containing frames	Not achieved by the current full annotated development preview; enclosure and release controls pending

R2 needs one frozen primary dimension per accepted class, or the same threshold ends up meaning different things to different people. We use equivalent-circle diameter for a compact pit, bubble, inclusion or contamination region; calibrated transverse width for a crack or scratch, with axial length reported separately; and the smaller calibrated bounding dimension for an abnormal-texture

patch. Class definitions and the ground-truth method are approved and frozen before data collection, so the 0.20 mm threshold cannot be quietly redefined once results are in. R3 works the same way: precision and recall must both pass for each of the three or more accepted classes, and we keep every class's event counts and confusion matrix so a macro or micro summary cannot bury a failing class. A pilot will indicate the minimum independent event count, but that count is frozen before the untouched test rather than picked afterwards.

R4 is measured per view rather than per specimen, and that has to be settled before the number means anything. One operator capture request runs the automatic cycle of Section 5.5, which drives both illumination branches in turn at one rotational index and so yields two views; R4 applies to each view separately, from the moment its branch begins its soft-start to the moment that view's preliminary result is displayed. A complete specimen is six rotational indexes under two branches — twelve views and one merged event set, as Section 5.2 sets out — and the specimen verdict is the merge of those twelve, carrying no one-second claim of its own. Keeping the soft-start and settling intervals inside the R4 budget rather than beside it is deliberate, because the operator waits through them: excluding them would let the requirement pass while the wait grew. It also makes the settling interval M5 measures at G2 a direct constraint on whether R4 can be met at all.

R4 is also where the design tension shows most clearly. A lower interim demonstration threshold may well help integration along, but it cannot quietly replace the one-second final target; moving that target takes a documented requirement change. If the final fixture misses it, the defensible responses are optimisation, a justified reduction in captured resolution, or formal renegotiation. Reporting only the runs that passed is not one of them.

4. Concepts and Selection

4.1 Selection method

We apply four mandatory gates before any preference scoring: availability of the issued primary imaging hardware, avoidance of an unapproved concept-specific primary-camera purchase, procurement timing, and semester deliverability. Each gate returns one of three outcomes, and telling them apart matters because they answer two different questions. A fail removes a concept from selection outright, however attractive it looks technically. A conditional pass keeps it in selection and names the evidence still owed. A deferral says only that the concept cannot be built and validated inside the window this report covers, which decides which concept carries the interim implementation rather than which concept is selected. Plan A is deferred on the last gate and selected on the strength of the other three, and that deferral is exactly why Plan B carries the interim build. These are team feasibility decisions for this proposal, not stakeholder-approved budget limits.

4.2 Plan A: close-range imaging of a held sample (selected concept)

Plan A connects ALC-IMX258-LY0001-ZB V1 camera module to the Pi over USB 2.0 and images a filament sample held in a rigid fixture. The supplier sheet identifies an IMX258 rolling-shutter sensor with 1.12 µm pixels, MJPEG/YUY2 output, autofocus or fixed-focus variants and several lens fields of view (Supplier product parameter sheet, 2026). The selected lens/focus option is not yet recorded, so close-range suitability remains a gate rather than a fact. For dimensional work, focus and exposure must be locked after setup because autofocus movement would change scale. The STM32 keeps the watchdog,commanded lighting and safe I/O functions, while the Pi performs defect classification, calibrated size estimation and approximate position estimation from the fixture datum and visible marker. Optical alignment, reflective cylindrical geometry, depth of field and single-view coverage all still need calibration.

The module is not a complete optical station. Its parameter table lists a main board, FPC-connected camera head and USB cable, not a stand, enclosure or controlled illuminator(Supplier product parameter sheet, 2026). The build therefore needs a rigid bracket and shroud, an indexed sample holder carrying a visible datum and view marker, diffuser and crossed-polariser parts for existing LED strip, and a calibration set that converts pixels into millimetres and tests localisation error.

4.3 Plan B: supplied-hardware distributed vision (interim implementation)

Plan B uses the university-supplied XIAO ESP32S3 Sense, Raspberry Pi 4 and STM32 Nucleo-L432KC (Monash University, 2026c). Its proposed chain is:

> Controlled optical head and rPLA filament sample -> ESP32S3 OV3660 acquisition -> 2.4 GHz Wi-Fi MJPEG -> Raspberry Pi 4 OpenCV, API/WebSocket, SQLite and GUI <-> STM32 watchdog, lighting and inhibited output interface

The OV3660 and ESP32 camera documentation supports the chosen acquisition stack (Espressif Systems, 2026; OmniVision Technologies, 2011; Seeed Studio, 2026), and the Pi and Nucleo documents support their planned processing and interface roles (Raspberry Pi Ltd, 2026a; STMicroelectronics, 2026). Plan B needs no new primary camera and builds on software and a live transport path that are already integrated, which is why it can carry the project while the Plan A hardware is in procurement. It also puts real limitations on display: JPEG artefacts, wireless delay or interruption, limited optical control, a fixed field that cannot be narrowed onto a 1.75 mm strand, and no guaranteed full-circumference view. Those constraints become test conditions and risk controls. Choosing Plan A removes the field and optical-control limits; it does not remove the rest.

4.4 Feasibility gates and technical comparison

Table 5 applies the mandatory gates first; Table 6 records the technical preferences afterwards.

Table 5. Mandatory feasibility gates.

Gate	Plan A	Plan B	Decision significance
Primary imaging hardware available, issued or approved for purchase	**Conditional pass**: the selected module is not issued or in hand	**Pass**: primary camera and computing platforms are listed as supplied. (Monash University, 2026c)	Plan A starts from an approved purchase instead of the issued baseline
No unapproved concept-specific primary-camera purchase	**Conditional pass**: A$40.00 is the team-reported item price, but still need delivered-cost evidence	**Pass** for primary imaging: no additional main camera purchase; common additions remain subject to approval	Approval removes the budget-authority objection; it does not by itself deliver hardware
Procurement/lead-time exposure	**Conditional pass with residual risk**: the camera is available through the team's own supply channel at a known price, but no order has been placed, no dated delivery	**Conditional pass with residual risk**: core platforms still need issue checks, while fixture, lighting and protection parts remain unfrozen	A supplier dispatch estimate is not a delivery commitment; neither concept may rely on an unverified arrival date

Gate	Plan A	Plan B	Decision significance
	commitment exists, and a retail product page is not durable evidence of either		
Semester implementation and validation	**Deferred**: the Plan A optical chain has not been integrated or calibrated within the window this report covers, so Plan B carries the interim build and Section 5.8 defines the migration	**Pass conditionally**: existing Plan B path can proceed to real-material tests	A deferral here bears on which concept can be built now rather than on which is selected: Plan A validation follows delivery, and Plan B interim validation remains conditional on material and fixture availability

A separate scoring exercise records technical preference, and it does not override those gates. Scores in table 6 are team-assigned comparative judgements from 1 (weak) to 5 (strong), not measurements.

Table 6. Weighted concept comparison.

Criterion	Weight	Plan A	Plan B	Evidence confidence	Basis of judgement
Defect-classification potential (R1, R3)	25%	5	3	Low (both): no rPLA accuracy data	Plan A offers a more configurable machine-vision optical chain; neither concept has rPLA accuracy data
Close-range imaging suitability (R2, R4)	20%	5	2	Low (A): field and working distance come from a retail listing	Adjustable narrow field with locked focus and exposure versus a fixed wide field and compressed wireless transport (Banggood, 2026; OmniVision Technologies, 2011)
Spatial-detail potential (R2)	20%	5	3	Low (A): sampling density is arithmetic (Sony Semiconductor Solutions, 2019); resolving power unmeasured	A field that can be narrowed onto a 1.75 mm strand versus the supplied OV3660 path
Raspberry Pi integration (R4, R6, R7)	15%	4	4	Low (A): a supplier claim. Medium (B): the path runs	USB video class versus supported network acquisition; the Plan A device's class compliance on the Pi is a supplier claim and is not yet verified

Criterion	Weight	Plan A	Plan B	Evidence confidence	Basis of judgement
Concept-specific cost (project constraint)	10%	4	5	Medium: one known price, five allowances	A$40.00 plus shipping for Plan A imaging, an approved purchase but the largest concept-specific line in the build; no new Plan B main camera
Procurement timing (schedule constraint)	10%	4	5	Low (A): no order, no dated commitment	Plan A depends on one delivery that has not been ordered or dated; Plan B core platforms listed as supplied, subject to issue check
Weighted total	**100%**	**4.65**	**3.35**	—	Comparative score only; the total ranks potential and does not certify it

Plan A leads the weighted matrix and remains the selected concept. so it is the selected concept. Plan B runs as the interim implementation on supplied hardware — which is what produced the evidence recorded in Sections 5 to 7 — and Section 5.8 defines the migration executed on delivery..

5. Functional Design and Specifications

The design splits the inspection task into two measurement paths, and they are deliberately unequal. The camera path carries the spatial evidence: where a visible candidate region sits and what shape it has. The auxiliary infrared path answers something much narrower — is material there, and is the optical head behaving the way it should? A candidate gets recorded only when acquisition is healthy. An automatic reject or stop request needs a good deal more than that: healthy acquisition, valid calibration, confirmed material presence, a validated event rule, and an output that has been enabled. That boundary is the heart of the design. A bright contour is not a defect just because software can draw a box around it.

The selected acquisition device is an integrated 4K inspection camera built on the IMX415 sensor (Banggood, 2026; Sony Semiconductor Solutions, 2019), connected to the Raspberry Pi 4 as a USB video-class device. Until that hardware arrives, acquisition runs on the university-supplied XIAO ESP32S3 Sense with OV3660 (Monash University, 2026c). The processing platforms do not change between the two: the Pi takes the non-deterministic vision and data services, and the STM32 Nucleo-L432KC takes time-critical I/O supervision. The one difference is where compression sits — the interim ESP32S3 compresses in the camera node and sends JPEG over the air, while the selected device compresses inside its own module and delivers frames over a local cable. Neither path is uncompressed, and the migration in Section 5.8 gains optical control and a shorter transport rather than raw pixels. Manufacturer documents back those platform capabilities (Espressif Systems, 2026; OmniVision Technologies, 2011; Raspberry Pi Ltd, 2026a; Seeed Studio, 2026; Sony Semiconductor Solutions, 2019; STMicroelectronics, 2026). What they cannot back is application performance, USB class behaviour on the Pi, or a finished Pi/STM32 deployment, and every measured value in this report came off the interim configuration.

5.1 End-to-end functional architecture

Figure 1. Proposed hardware, data and inhibited-action architecture, showing the selected Plan A acquisition front end and the interim Plan B front end that feeds the same downstream chain.

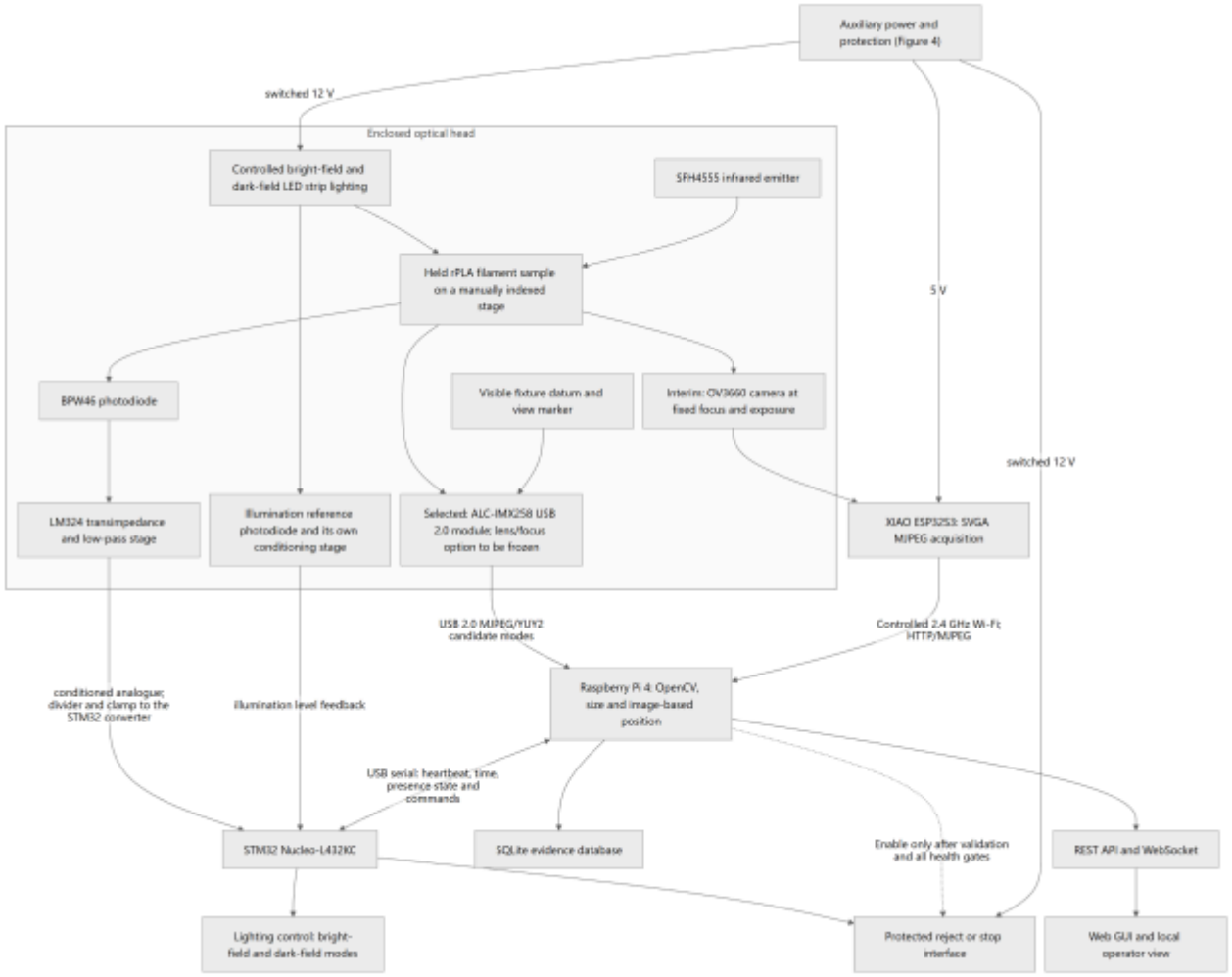


Figure 1 puts the vision and service functions we developed on Windows into the Pi role they are meant to occupy, and separates them from the action authority the STM32 is meant to hold. The arrows describe the target integration baseline, not wiring that exists. Both acquisition front ends hand frames to the same source-independent contract, and only one runs at a time. The visible fixture datum and view marker read by the Pi form the complete position-reference path.

The diagram also marks the action boundary. Raspberry Pi software may propose an event; the STM32 owns the physical output and withdraws permission the moment heartbeat, lighting or communications health is lost. This is a safety-oriented laboratory control architecture. It is not a safety-certified machinery function, and that difference is not a matter of wording. Connecting any of it to production equipment later would need a machine risk assessment and a safety-function design against the relevant framework, ISO 13849-1 included, rather than a prototype watchdog carrying that weight alone (International Organization for Standardization, 2023). OT network segmentation and least-functionality principles would apply as well (Stouffer et al., 2023).

Every requirement below still carries the `PROPOSED_REQUIREMENT` label. The project brief establishes the need for laboratory-scale, online surface-quality monitoring (Monash University, 2026d), and the discipline guidance requires measurable specifications with a verification route (Monash University, 2026a). Neither of those amounts to a stakeholder approving our numbers, and

no record supplied to this report shows that anyone has. Table 7 assigns each requirement to functions and a verification route, and leaves the gap visible rather than papering over it.

Table 7. Functional allocation and verification status of the proposed requirements.

ID	Proposed target	Functional allocation	Verification and present status
R1	Detect at least three visible surface-defect classes.	Optical head; OpenCV/event classifier; labelled database.	Independent real-rPLA event test. `PENDING_REAL_MATERIAL`; current boxes are candidates only.
R2	Demonstrate the frozen class-specific characteristic dimension at or above 0.20 mm.	Fixed optics, pixel scale and class-specific event measurement defined in Section 3.2.	Calibrated reference features at final working distance. `PENDING_REAL_MATERIAL`.
R3	For every accepted R1 class meeting R2, achieve event-level precision and recall of at least 90% per class.	Event logic, ground truth and evaluation scripts.	Held-out spool/segment/event evaluation; report per-class counts, confidence intervals, F1 and macro/micro summaries. `PENDING_REAL_MATERIAL`.
R4	Return a preliminary result within 1 s, measured per illumination branch from the start of its soft-start.	Selected chain: camera, USB transport, newest-frame policy and Pi vision. Interim chain adds the Wi-Fi leg.	Timed capture-to-display runs on the final fixture. The recorded 10.77 FPS SVGA integration result bounds interim preview responsiveness but is not this measure; latest-frame and drop instrumentation are now wired into the resident service but remain unmeasured on the Pi.
R5	At P95, assert the alarm within 500 ms of the first frame in which a marked defect is clearly visible with illumination health already confirmed.	Timestamped event tracker, confirmation rule, API/UI and STM32 output request.	One video carrying the illuminated field, a millisecond reference and the GUI/output indication. Existing brightness-step timing is only `LIVE_INTEGRATION_ONLY` and is not this measure.
R6	Store type, size, confidence, evidence image, specimen and view identifiers, time and batch identifier with required calibration provenance.	`DetectionRecord`, SQLite, image store, calibration manifests and export.	Schema/file round-trip test, then calibrated record audit. Current generic calibration linkage and live image persistence are incomplete; physical fields remain null.

ID	Proposed target	Functional allocation	Verification and present status
R7	Operate for 30 min with confirmed-event record loss below 1%.	Resident service, SQLite transaction path, durable image store and health log.	Reconcile frozen expected event IDs to complete durable rows/images/exports using the Section 3.2 formula; report acquisition/processing frame drops separately. Not yet tested.
R8	Enter a safe hold or fault state after camera, light or processor failure.	Pi health gate, STM32 watchdog and protected output.	Hardware fault injection with the final wiring. Software/network evidence exists, but safe I/O remains pending.
R9	Store and serve only the enclosed filament-tunnel view, locally by default.	Shrouded optical head, release crop/mask and Pi-local database.	Audit every final-evidence frame and storage/access configuration for zero outside-tunnel/person content. Current development endpoints serve a full annotated frame rather than an ROI crop; R9 is not achieved (Australian Government, 2025; Office of the Australian Information Commissioner, 2026).

5.2 Optical head and camera specification

The selected camera is an integrated ALC-IMX258-LY0001-ZB V1 module.Its supplier table lists a 1/3.06-inch IMX258 CMOS sensor, 1.12 µm pixels, rolling-shutter readout, USB 2.0, MJPEG/YUY2 formats, 5 V operation at 120–250 mA, and candidate modes including 3840 × 2160 MJPEG at 30 frames/s and 3840 × 3104 MJPEG at 15 frames/s (Supplier product parameter sheet, 2026). It also lists autofocus from 4 cm to infinity, fixed-focus and macro variants, and lens fields of view from 65° to 165°.

What matters here is scale, and scale is arithmetic. For a horizontal pixel count N_h spread over an imaged field width w_f, the optical sampling density s and the pixel span of a feature of width d are

$$s = \frac{N_h}{w_f}, \qquad n_d = s\,d.$$

For planning only, a negotiated 3840-pixel-wide mode combined with a measured 40 mm field would give (s=96) px/mm, about 168 pixels across a 1.75 mm strand and 19 pixels across a 0.20 mm feature. The field is an assumed commissioning target, not a camera specification. The exact lens, working distance, focus state and distortion determine whether it can be achieved. Sampling density is not resolving power, so G3 images a resolution target and measures depth of field instead of inferring either from pixel count. The rolling shutter adds another condition: the specimen and fixture must settle before exposure, and captures made during handling or vibration are invalid. Focus, exposure and white balance are locked after setup if the delivered controls permit it; otherwise the module does not pass the measurement gate. The millimetres-per-pixel scale is fixed only by calibration, and the supplier's frame-rate claims do not establish R4 because Pi-side capture, decode, inspection and storage are still unmeasured.

The interim configuration uses the supplied OV3660. It can expose larger frames, but what the project needs is a stable operating point, not the biggest mode in the datasheet. SVGA 800 × 600 is fixed as the interim live profile; VGA 640 × 480 stays available as a manual recovery option, and XGA or anything larger falls outside the selected operating configuration. The active normalised

ROI maps at SVGA to `(254, 47, 195, 527)`. Storing the ROI in normalised coordinates is deliberate: it survives the change of sensor and frame size at migration. Its position stays provisional until the fixture and the filament settle the true optical axis. OV3660 readout, JPEG compression and wireless transport can each introduce timing or block artefacts, so exposure duration, illumination mode and candidate persistence have to be tested together rather than one at a time (Espressif Systems, 2026; OmniVision Technologies, 2011). The wireless leg and the fixed wide field are the specific limits the selected configuration removes; compression is not one of them, because the selected module also encodes inside its own housing.

The proposed optical head is designed to provide stable repeatable image-acquisition conditions. A rigid camera bracket maintains a consistent working distance and view angle, while a matte-bleck indexed sample holder positions the filament on the optical axis and allows controlled rotation between different views. The filament is secured outside and region of interest so that only the inspected surface appears in the captured image. A matte background and external shroud are also used to minimise reflections and reduce interference from ambient light.

After calibration, the working distance, field of view, focus and exposure settings will be fixed and recorded, the lighting system can switch between diffuse bright-field illumination, which is intended to improve colour and texture contrast, and low-angle dark-field illumination, which uses shadows to emphasise surface relief and small irregularities. Both illumination modes will be tested under the same optical geometry to determine which provides the clearest and most repeatable representation of each defect type. These lighting modes are experimental conditions rather than evidence that a particular defect can already be detected. Variations in illumination and viewing angle will therefore be included in the validation dataset (Carvalho et al., 2024; Heckler-Kram et al., 2026; Wang et al., 2024).

One camera sees only one arc of a round object. Holding the sample makes coverage mechanical: the holder is manually indexed through six planned 60° sectors, each identified by a visible marker, and each sector is captured under both illumination modes. A specimen therefore produces twelve planned images and one merged event set. The ±45° usable-arc assumption and 30° overlap are provisional and must be checked on the delivered lens. The head also carries a scale target plus dark and white references that can be inserted without moving the camera. None of this supports a physical-resolution or full-coverage claim until tested.

The Raspberry Pi estimates position from the detected fixture datum, candidate coordinates and calibrated image scale. Let (u_0, v_0) be the detected fixture datum, (u_c, v_c) the centre of a candidate box, and (s_x, s_y) the calibrated scale in millimetres per pixel. The proposed transform is

$$x_{mm} = s_x(u_c - u_0); \quad y_{mm} = s_y(v_c - v_0)$$

For this filament fixture, the operator-facing position is the approximate axial distance x_{mm} from the datum plus the visually decoded view sector. location_mm, location_uncertainty_mm and view_sector remain null if the datum, view marker or calibration is invalid. This does not reconstruct the hidden circumference or track a moving production line.

5.3 Auxiliary photodiode chain

The SFH4555 sends infrared light across the filament, and the BPW46 receives the light and produces a photocurrent. This helps the system check whether the filament is present and whether the light path is clear. The LM324 converts the photocurrent into a voltage and reduces high-frequency noise before the signal is read by the STM32 ADC. The first estimate of the transimpedance resistance is:

$$R_f = \frac{V_{span}}{I_{PD}},$$

where I_{PD} is the photocurrent measured in the final setup and v_{span} is the usable output voltage range. With a feedback C_f,the low-pass cutoff frequency can be estimated by:

$$f_c = \frac{1}{2\pi R_f C_f}.$$

The final value of R_f, C_f, the LED current and the detection thresholds cannot be decided until the optical head is built and tested. M3 will choose the gain so that the measured signal uses about 10-80% of the ADC input range without clipping. The value of C_f will then be chosen to reduce high-frequency noise while keeping the response ai least five times faster than the fastest change in filament presence.

The LED current will be kept below the manufacturer's continuous current limit and checked again at the measured enclosure temperature. At G3, the circuit will be tested under five conditions: dark, clear path, normal filament, blocked path and misaligned filament. The results will be used to set the detection threshold with a suitable safety margin. The auxiliary signal can be used to check filament presence or show a problem with the optical path. However, it cannot locate scratches, identify contamination or measure the size of a defect.

The analogue signal is connected to the STM32 instead of the Raspberry pi. The STM32 Nucleo-L432KC already has a built-in ADC (STMicroelectronics, 2026), so the MCP3208 will be kept as a spare and will not be used in the final design. Using another SPI ADC is not necessary because there is only one auxiliary analogue channel.

Connecting the signal directly to the STM32 also allows the STM32 to check the filament presence and optical path condition. The result is then sent to the Raspberry Pi through the existing Pi-STM32 communication, so the health gate in figure 2 can still use the same input.

Before the final connection is made, M3 will check the ADC resolution, input voltage range and available pins using the device datasheet. The LM324 is not a rail-to-rail device, so its output voltage also needs to be checked when it is powered by the regulated 5V supply. Both normal and fault conditions will be considered.

M3 will use suitable gain and bias settings, an input series resistor, and a clamp/divider circuit to keep the signal within the safe input range of the STM32 ADC. The minimum and maximum voltage and the protection current will also be measured before connecting the circuit to the STM32. The final component value will be decided during G2/G3 testing. Raspberry Pi GPIO and SPI use 3.3V, so 5V analogue or digital signals must not be connected directly to either controller. The official component information used for these limits is given in Section 8.

One detector cannot be used for every purpose. The lighting control in Section 5.5 needs to measure the actual light output from the optical head, but the current infrared channel cannot provide this information by itself. The silicon photodiode can detect both visible and near-infrared light. This means its reading may change when the filament color changes, then the holder moves to another position, or when the lighting level changes. Therefore, one photodiode reading cannot clearly show what caused the change.

To avoid this problem, two separate measurements are used. Filament presence and optical path condition are measured during the dark period of the capture cycle, when both visible light sources are turned off. This allows the channel to mainly measure the infrared signal. At G2, M3 will check

whether turning on either visible light source affects the reading. If it does, an infrared-pass filter will be added to the detector.

The lighting control uses a second photodiode with its own signal conditioning circuit and ADC channel. This photodiode is pointed at the diffuser output instead of the filament, so it measures the light output rather than changes in the sample. The calibration in Section 7 also keeps these two measurements separate. The five filament presence conditions are measured by the infrared channel, while the light level of each lighting branch is measured by the reference detector. This prevents the same signal from being used for two different purposes. The reference detector cannot detect changes in the position or geometry of the setup, so the placement tolerance will still be tested separately in Section7.

5.4 Raspberry Pi processing, provenance and storage

The planned Raspberry Pi deployment preserves the source-independent contracts established in Week 3 rather than replacing them. Frame Packet stores the image, Frame Detection Record stores the candidate label, pixel bounding box, pixel area, algorithm score, provisional grade, and batch context. The SQLite database includes tables for batches, detections, system events, and calibrations. The service also enables SQLite foreign-key enforcement, preventing a detection from referring to a batch that has not been opened.

These boundaries support auditability when the algorithm is changed in the future, and they are consistent with the role of edge computing in recent industrial vision research (Katsigiannis & Mykoniatis, 2024). However, the schema is not yet complete. It currently uses one generic calibration_id, while the calibration table records only the optical scale in pixels per millimetre and an evidence path. It does not record the illumination mode, rotational index, or resolution-target evidence that supports a size claim. Therefore, physical size measurements cannot yet be fully traced. The released design should include a calibration record that links the optical scale to the capture conditions under which it was measured.

Figure 2. Health-gated candidate-processing and evidence-record flow.

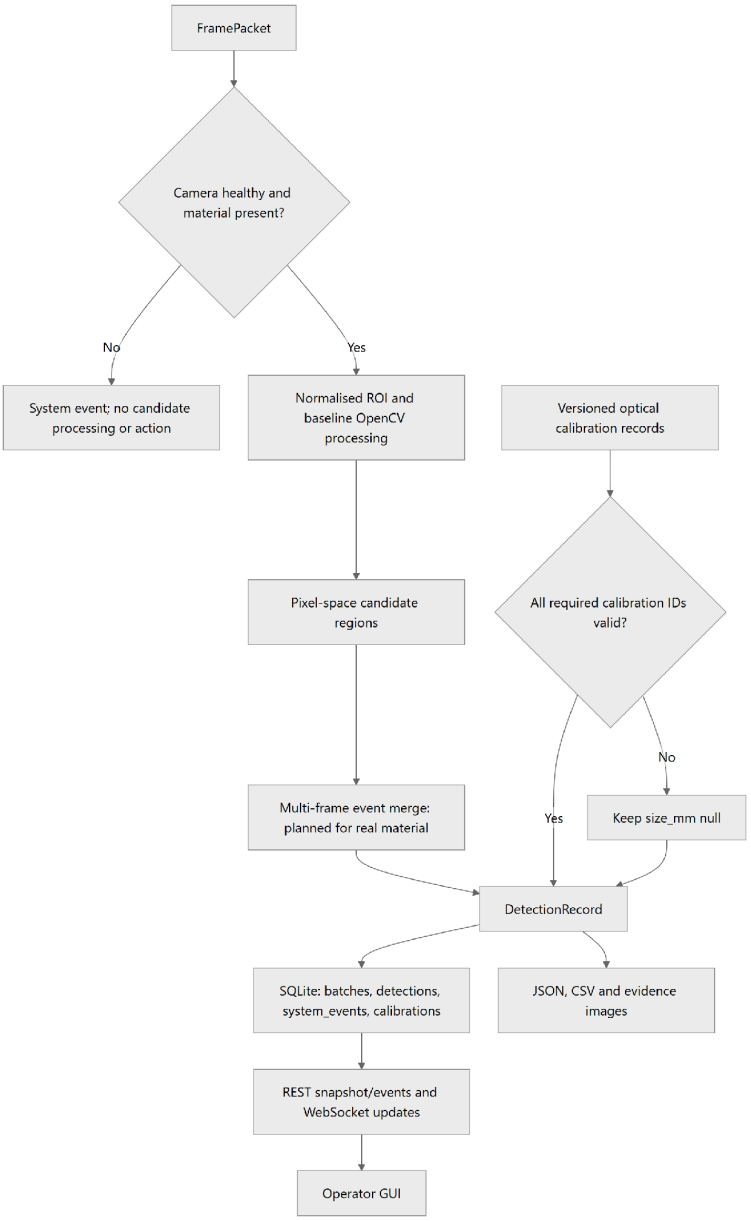


Figure 2 shows where health, calibration and provenance step in to stop a pixel-space candidate from picking up physical meaning it has not earned.

The Windows development baseline uses a configurable threshold of 145, a minimum area of 120 px, and an enabled material-presence gate (ENG5105 Project Team, 2026a, 2026i). These remain prototype parameters. The gate currently requires a dark target across 80% of the ROI height. This prevents a nearby hand or small background object from starting inspection, but it must be returned once rPLA is placed before the camera. The one-slot Latest Frame Buffer and the 1 s Stale/2 s Disconnected monitor were initially OFFLINE_FAULT_INJECTION_ONLY primitives; the resident service now creates both. Acquisition and inspection use separate threads. The reader publishes each frame to the one-slot buffer. Backlogged frames are discarded and counted, while the inspection thread assesses health from the age of its newest frame, rather than from whether the latest read finishes within the configured 3000ms timeout. This distinction is important: a frame older than the stale window lowers the service state even when reads continue to succeed. Inspection also requires three consecutive frames before it resumes. The dropped frames field now records buffer discards rather than a constant zero, and each snapshot includes frame age_ms with the configured thresholds. These findings come from threaded unit tests on the development host. They do not prove Pi deployment, blacklog suppression under real network load, or R4,R7, and R8,

which still require the final fixture. Mult-frame merging creates event candidates from repeated frame candidates only after representative material is available.

The current model-level rule is strict: without a generic calibration_id, size_mm remains null. However, this safeguard does not prove provenance. The resident live service leaves image_path and calibration linkage null. And the calibration table cannot identify the optical_path and calibration linkage null, and the calibration table cannot identify the optical-scale evidence required for a size claim. Synthetic records are marked OFFLINE_SYNTHETIC_ONLY, while live camera candidates are marked LINE_INTEGRATION _ONLY. Neither supports a physical claim. Before R6 passes, the release schema must require the relevant calibration IDs or a composite-manifest hash, together with a durable evidence-image path.

Illustrative input and output (not a test result)

The figure 3 only demonstrates the intended interface. Every value is assumed, the calibration identifier is fictional, and ILLUSTRATIVE_ONLY prevents the record from entering validation evidence.

Figure 3. Illustrative input and output


```
{
  "evidence_status": "ILLUSTRATIVE_ONLY",
  "sample_id": "RPLA-DEMO-001",
  "view_marker": "V3_120deg",
  "illumination_mode": "low_angle",
  "frame_mode": "3840x2160_MJPEG_ASSUMED",
  "calibration_id": "CAL-A-DEMO",
  "datum_visible": true,
  "assumed_scale_mm_per_px": 0.010,
  "candidate_bbox_px": [1860, 940, 31, 108]
}
```

```
{
  "evidence_status": "ILLUSTRATIVE_ONLY",
  "message": "Surface crack detected; characteristic width approximately 0.31 mm; located approximately 18.6 mm from the left fixture
datum in the 120-degree view sector.",
  "defect_class": "surface_crack",
  "characteristic_size_mm": 0.31,
  "approximate_location": {
    "axial_mm_from_left_datum": 18.6,
    "view_sector": "120deg",
    "method": "Raspberry Pi image coordinate transform"
  },
  "decision": "REVIEW"
}
```


5.5 STM32 supervision and safe I/O

The proposed STM32 firmware will sample the auxiliary optical channel, debounce capture requests, monitor the Pi heartable, control two illumination branches through rated and protected transistor drivers, report the active mode, and operate the reject/stop demonstrator output (STMicroelectronics, 2026). However, none of these functions has yet been tested on physical STM32 wiring.

The STM32 has a limited but essential supervisory role; it is more than a message-forwarding link. It handles hardware-critical decisions independently of the general-purpose OS and web Stack, including material presence, optical-path health, the energised illumination mode, Pi availability, and permission for physical output. Because lighting affects surface-quality assessment (Monash

University, 2026d), the bright-field and low-angle dark-field branches are controlled separately as commanded states. The active mode is returned in the existing Pi–STM32 frame, and capture is rejected when the reported and requested modes disagree. Illumination therefore becomes part of the evidence record. The camera's built-in illuminator remains off; otherwise, the recorded mode would not represent a commanded condition. Before release, G2 must verify driver levels, switching, branch protection, and mode reporting.

Illumination is automatic. The operator presses capture once for each specimen position, while the controller runs both modes in sequence and maintains each at a measured level rather than an assumed brightness. This produces consistent images and ensures that the recorded mode and level reflect actual hardware behaviour. Manual lamp adjustment would make each capture a different measurement

A capture cycle runs as follows.

1. The request arrives on the debounced input.
2. Both branches are de-energised, and the infrared channel of Section 5.3 is sampled in that dark window; the request is accepted only if material presence and optical health hold.
3. The bright-field branch is ramped from zero across a soft-start interval to its calibrated drive level.
4. The controller trims that level until the reference detector of Section 5.3 reads inside the band recorded for the branch at calibration, then reports that capture may proceed.
5. The Pi captures, and the record stores the branch, the drive level actually used and the measured reference value.
6. Steps 2 to 5 repeat for the low-angle dark-field branch.
7. Both branches return to off, which is also the safe state.

The trim in step 4 is what makes the illumination repeatable, and how it is bound matters more than how it is tuned. Drive level is the actuator and the reference detector is the feedback, so the loop regulates the light the head emits rather than the current the driver delivers. Calibration seeds it: with a white reference in the field, G3 records the drive level and the reference value that give a correct exposure for each branch, and those two numbers become the set point and the acceptance band. Correction is deliberately slow, because what it exists to absorb is slow — emitter warm-up, thermal droop and ageing — and a faster loop would chase image content instead.

Authority over that correction is limited, and the limit is what carries the evidence argument. The controller may move the drive level only inside a window around the calibrated value. Needing to leave that window to reach the set point means something physical has changed, a degraded emitter or a clouded diffuser, so the controller raises an optical-health fault rather than compensating for it. An unbounded loop would keep the picture looking correct while the optical head quietly degraded, which is precisely the failure this design cannot afford. Camera exposure stays locked for the same reason: two loops competing over the same brightness would leave neither the image nor the record meaning anything.

One manual path stays open. During commissioning M5 can set a drive level by hand to explore geometry, and any capture taken under a manual level is flagged in the record so that it can never be counted as calibrated evidence. Six quantities are frozen before any of this is released, and none of them can be taken from a datasheet: the soft-start interval, the calibrated drive level and acceptance band for each branch, the settling interval, the correction limit and the loop rate. M5 measures the strip's rise to steady output and the camera's exposure convergence at G2 and sets the settling interval from the slower of the two with margin; G3 fixes the presence thresholds from the measured five-state distributions, and the illumination set points and bands from the white-reference exposures. Table 11 allocates the lighting 8.00 W normal and 12.00 W peak, so holding a branch

energised through a long session is a thermal question rather than an electrical one, and the measured enclosure temperature decides whether a cool-down between captures is required.

Figure 4. Proposed acquisition-health states and output-inhibition behaviour.

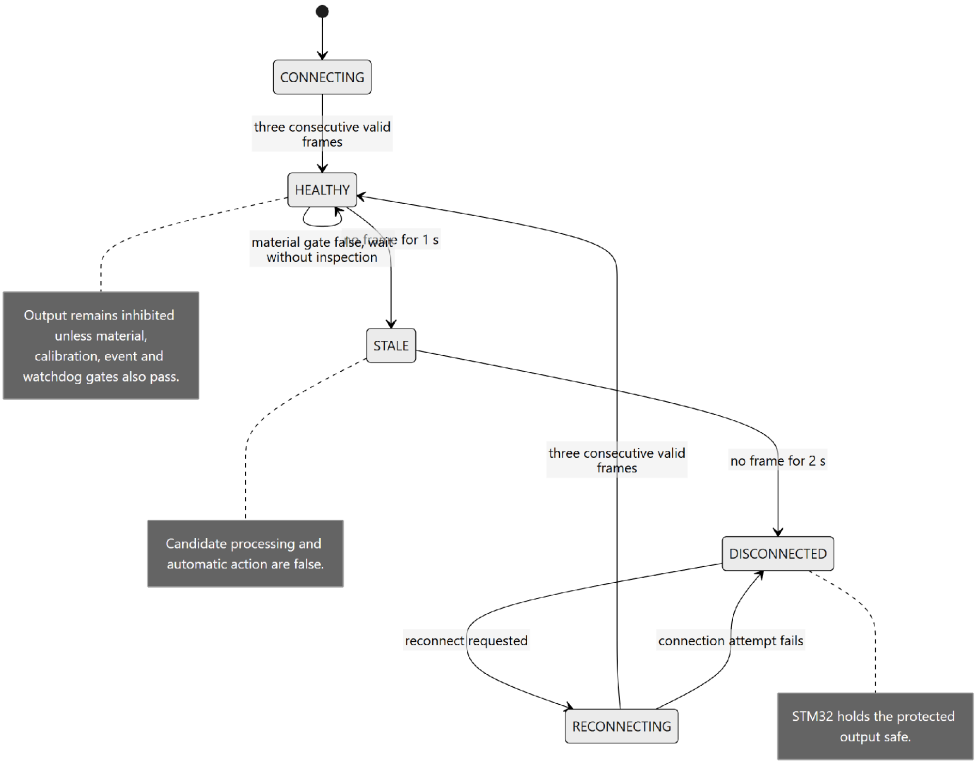


Figure 4 defines what counts as a qualified recovery, and which states must keep candidate processing or physical output inhibited. The resident Windows service now enforces the 1 s `STALE` and 2 s `DISCONNECTED` transitions and the three-consecutive-frame recovery rule, judging health from the age of the newest buffered frame rather than from whether a read returned inside its configured 3000 ms timeout, as Section 5.4 records. That is development-host evidence from threaded unit tests. It does not establish Pi deployment, behaviour under real network load, or any of the R4, R7 and R8 results, so the state timing in the figure stays a design awaiting verification on the target hardware.

The released Pi service must not replay old frames. It will require three consecutive valid, age-bounded frames before processing resumes, and a Pi restart or a heartbeat loss will clear output permission. After a latched hardware fault, re-enabling any automatic demonstrator will take manual acknowledgement. If the web interface fails while the STM32 is still alive, the operator loses visibility, but the proposed watchdog and output inhibition stay local to the control path. These are target behaviours, pending live-service integration and the final wiring tests.

5.6 Communications, API and user interface

Table 8 separates transport that is implemented from hardware interfaces that are only proposed. Table 9 then fixes the Windows development API behind the present GUI evidence.

Table 8. Subsystem communications and integrity rules, covering the selected and interim acquisition chains.

Interface	Payload and present/planned mode	Integrity, timing and recovery rule
Inspection camera → Pi (selected)	Supplier-claimed USB 2.0 MJPEG/YUY2 modes, including candidate 3840 × 2160 and 3840 × 3104 profiles (Supplier product parameter sheet, 2026); no wireless leg.	`PROPOSED_INTERFACE`: Linux enumeration, negotiated mode, control lock, USB bandwidth, cable seating and capture-stall behaviour must be confirmed on the Raspberry Pi and fault-injected at G5 before migration is accepted.
OV3660/ESP32S3 → Pi (interim)	HTTP multipart MJPEG over controlled 2.4 GHz Wi-Fi; SVGA 800 × 600. Replaced by the USB link on migration.	Resident service is the only direct MJPEG client. Read/open timeout and reconnect logic gate processing; rate is measured, not assumed.
Pi ↔ STM32	Proposed USB serial baseline: 115200 bit/s, 8-N-1; framed monotonic sequence, millisecond timestamp, material-presence state, active illumination mode, health bitmask, lighting command and output request.	`PROPOSED_INTERFACE`: CRC-16 per frame and 10 Hz heartbeat; three invalid/missing heartbeats or more than 300 ms without a valid frame inhibits output. M5 freezes the framing/timing after G2 bench checks, and G5 verifies fault response.
Capture request → STM32	Operator capture button as a debounced digital input with a hardware pull-up, read alongside the sample-presence state derived from the optical channel.	`PROPOSED_INTERFACE`: a capture is accepted only when presence, optical health and the reported illumination mode all agree with the request; debounce interval, accept/reject rule and input protection are frozen after G2 bench checks.
Optical channel → STM32	Proposed conditioned analogue input sampled by the microcontroller's converter, timestamped and reported in the existing Pi–STM32 frame; initial bench rate 200 samples/s per active channel.	`PROPOSED_INTERFACE`: for the shortest measured presence transition t_{min}, require $f_s \geq 10/t_{min}$ so at least ten samples span it; stale/out-of-range values create an optical-health fault. M3 sets thresholds from the five-state distributions at G3.
STM32 → local LCD	I2C summary of system health and safe-output state.	Diagnostic only; it does not override the web/database record.
Pi → browser	REST and WebSocket over the local Ethernet/Wi-Fi network.	Local-by-default access; the UI must show disconnection and last update rather than silently freeze.
Future plant link	MQTT or Modbus TCP/RTU is an option, not part of the validated baseline.	Requires an interface-control revision, OT security review and separate acceptance (ISO/IEC/IEEE, 2023; Stouffer et al., 2023).

The implemented Windows development service listens on port 8765 (ENG5105 Project Team, 2026e). Its current API contract is explicit:

Table 9. Implemented development-service API contract.

Method and endpoint	Response or action	GUI use
`GET /api/health`	Acquisition, processing, gate and evidence state, including the development field `automatic_actions_allowed`.	Health banner and diagnostics; this field is not actuator authorisation.
`GET /api/snapshot`	Current summary plus candidate records.	Initial page state and resynchronisation.
`GET /api/events?limit=`	Bounded event history for the active batch.	Event table and export preparation.
`POST /api/events/delete`	Development-only hard deletion of selected database rows. It now writes a `DETECTION_RECORDS_DELETED` event naming the requested identifiers, but still records no operator identity, reason or tombstone.	Traceable clean-up only; disable before evidence release and replace with role-identified quarantine/tombstone handling.
`GET /api/frame.jpg`	Latest annotated JPEG, or 503 before a frame is available.	Single-frame view.
`GET /api/stream.mjpg`	Shared annotated MJPEG output.	Live view without another direct camera client.
`POST /api/control/reconnect`	Requests camera recovery.	Manual recovery control.
`WS /ws`	Pushes changed snapshots.	Low-delay health and event updates.

These endpoints are Windows-host implementation evidence. Deployment and start-up persistence on the Raspberry Pi, the USB serial contract and physical STM32 I/O are all still plans. One name in the present service is actively misleading. `automatic_actions_allowed` is no more than a development readiness Boolean computed from `camera_state == HEALTHY` and the material-presence gate. It does not check calibration, event validation, STM32 heartbeat, fault latches or operator authorisation, and no actuator is connected to it. It must be renamed before release. The target interface should expose a separate `safe_output_authorized` that goes true only when frame data is not stale, optical calibration is valid, the event rule is validated, the reported lighting mode matches the requested one, the Pi–STM32 heartbeat and watchdog are live, the operator has explicitly enabled output, and no fault is latched. Until every G5 release condition passes — physical fault injection, a confirmed safe default level and owner sign-off — that value stays false and physical output stays inhibited. The hard-delete endpoint now writes an audit event naming what was removed, but an irreversible delete still contradicts the proposed append-only evidence governance and has no place in an evidence-release configuration. Before acceptance it should be disabled, or replaced by a controlled soft delete carrying identity, time, reason, retention policy and an append-only audit record. Existing evidence must never disappear without a trace.

The GUI should let an operator understand the whole chain at a glance: camera and material-gate health, evidence label, live ROI, candidate overlays, frame rate, post-decode-to-GUI latency, batch and calibration identifiers, Pi/STM32/lighting states, event history and export. Before calibration it shows pixel measurements. Physical size and position stay blank, and candidate status must never

be dressed up as a validated quality grade. The local LCD carries only a small health summary, so that losing the browser does not look like losing the controller.

5.7 Power budget and protection

The design keeps the computing system separate from the lighting and other switched loads. Table 10 shows the power allocation for the 5.1 V computing system, including the selected USB-powered camera. Table 11 shows the power allocation for the 12 V auxiliary system, including the interim camera, analogue circuit, lighting and demonstrator. These values are used for initial power planning and are not measured power consumption. The camera power is currently based on the supplier's stated range of 5V and 120-250 mA. This will be checked using the measured start-up and steady-state current at G2 (Supplier product parameter sheet, 2026). M3 will also measure the power use of each part of the system before the fuse and wire sizes are finally selected.

Table 10. Proposed 5.1 V Raspberry Pi-domain power allocation.

5.1 V Raspberry Pi domain load	Normal	Peak
Raspberry Pi 4	7.50 W	12.50 W
STM32 Nucleo via USB	0.25 W	0.50 W
ALC-IMX258 USB camera, selected chain	0.60 W	1.25 W
Domain total	**8.35 W**	**14.25 W**

A 5.1 V, 3 A supply provides $P_{available} = 5.1 \times 3 = 15.3\ W$. The allocated peak margin is

$$M_{Pi} = \frac{15.3-14.25}{15.3} \times 100 = 6.9\%.$$

Table 11. Proposed 12 V auxiliary-domain power allocation.

12 V auxiliary-domain load	Normal	Peak	Path
ESP32S3 Sense + OV3660	1.50 W	2.50 W	12 V to regulated 5 V
IR/analogue/LCD electronics	0.40 W	0.80 W	Mixed regulated rails: LM324/auxiliary devices as confirmed; conditioned output clamped to the STM32 analogue input range
Controlled lighting	8.00 W	12.00 W	12 V branch
Reject/stop demonstrator	0 W	5.00 W	Protected switched branch; calculation treats 5.00 W as a 12 V-side allocation
4K inspection camera (selected chain)	—	—	Its own mains adapter, supplied with the package; outside this domain, and carrying data to the Pi over USB only

Assuming 85% conversion efficiency for the 5 V loads,

$$P_{12V,normal} = \frac{1.50+0.40}{0.85} + 8.00 = 10.24\,W,$$

$$P_{12V,peak} = \frac{2.50+0.80}{0.85} + 12.00 + 5.00 = 20.88\,W.$$

A regulated 12 V, 5 A source supplies 60 W, leaving

$$M_{12V} = \frac{60-20.88}{60} \times 100 = 65.2\%.$$

The selected camera does not appear in either domain total, and that is a claim the incoming check has to confirm rather than an omission. The package is listed as including a power supply (Banggood, 2026), so the design assumes a mains adapter feeding the camera directly, with the USB link carrying data alone. If the camera is found to be USB-powered, its power must be added to the 5.1 V system. A Raspberry Pi 4 USB port rated at 900 mA can provide up to 4.50 W. This would increase the peak power in Table 11 from 13.00 W to 17.50 W, which is higher than the 15.30 W available from the current supply. In this case, a larger 5.1V supply or a powered USB hub will be needed before the system is connected. The grounding and protection between the camera adapter and the auxiliary system will also need to be clearly defined.

The current peak power calculation is only an initial estimate and is not the final electrical design. The 20.88 W value assumes the demonstrator's 5.00 W is already expressed at the 12 V source; if the selected device draws from a converted 5 V rail instead, its peak has to be divided by the measured converter efficiency and the total recalculated before G2. Week 5+ tests must capture start-up inrush, steady current, Pi undervoltage logs and regulator temperature before fuse, conductor and connector ratings are frozen. Each power branch will have its own fuse, reverse-polarity protection and labelled isolation. Inductive loads will use flyback protection, and the lighting will be controlled by a suitable transistor/MOSFETdriver. Signal grounds will connect at a controlled ground point. The raspberry pi power supply will not be used to power the demonstrator branch.

Figure 5. Auxiliary power and protection domain, showing the branches the 12 V source feeds and where protection and switching sit.

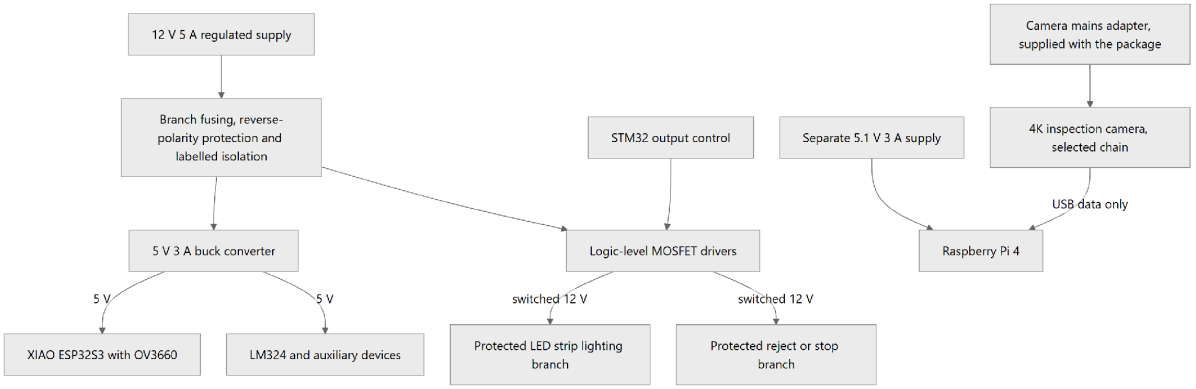


Figure 5 shows the computing power supply and the auxiliary power supply as separate parts of the system. The selected camera also has its own power adapter, which is shown separately from the 12 V supply and the Raspberry Pi supply. This makes the camera power arrangement clear in this diagram.

The Raspberry Pi supply is not used to power any switched loads. Each 12 V branch has protection before it is connected to the load, and the two switched branches use logic-level drivers controlled by the STM32. The auxiliary analogue circuit uses the microcontroller supply as its reference instead of the 12V supply. This helps keep the signal within the ADC input range described in

Section 5.3. The final component value, fuse ratings and wire sizes will be selected after the measurements at G2.

5.8 Migration from the interim configuration to the selected concept

We chose the interim path so that migration would be a front-end replacement rather than a redesign. Everything from the frame contract downwards is shared: the source-independent contracts recorded in Appendix A (ENG5105 Project Team, 2026a) may not be reached into by any camera adapter, the acquisition adapter is injected through a `capture_factory`, and the inspection thread receives frames without knowing where they came from. That separation already exists in the current code, and it is the reason the selected concept can be adopted without reopening the requirement set, the supervision layer or the evidence schema.

Table 12 states what changes, what is preserved, and what evidence has to be re-established. The last column is the one that matters. Migration replaces the measurement instrument, so every value describing optical or timing performance dies with the change and has to be measured again on the new chain. No interim figure crosses over.

Table 12. Migration from the interim OV3660 path to the selected close-range USB path.

Layer	Change on migration	Preserved	Evidence status after migration
Camera and transport	The ALC-IMX258 USB 2.0 module replaces OV3660 over 2.4 GHz MJPEG; the exact lens/focus option and negotiated USB mode are frozen	Enclosure, sample holder, matte background, lighting modes	All acquisition timing and image-quality evidence is void and must be re-recorded
Acquisition software	A Linux camera adapter replaces the MJPEG client behind capture_factory; the one-slot latest-frame buffer and frame-age monitor are retained and re-tuned	`FramePacket` inspection thread, health state machine, drop instrumentation	Enumerate and stress-test the delivered module on the Pi; re-run contract tests with the new adapter
Optical calibration	New scale, distortion, fixture-datum and view-marker calibration for the delivered sensor/lens/working distance; a new manifest is issued	Normalised ROI definition and provenance rules	Previous calibration is retired; size and millimetre location return to null until the new manifest passes
Processing, storage and interface	Extend the schema with illumination mode, view sector, datum ID, coordinate uncertainty	OpenCV baseline, `DetectionRecord`, REST/WebSocket API, GUI, export	Contract tests re-run; pre-extension records cannot be promoted to physical evidence

Layer	Change on migration	Preserved	Evidence status after migration
	and evidence-image provenance		
Supervision and safe I/O	No change required	STM32 watchdog, heartbeat, lighting control, capture gating, inhibited output, G5 release gate	Fault-injection evidence must be repeated on the migrated wiring before the G5 gate is re-cleared
Requirements	No change to the retained targets	Requirement identifiers, verification routes, acceptance rules	R2, R3, R4 and R5 results revert to `PENDING_REAL_MATERIAL` and are re-measured on the selected chain

Three conditions apply before any Plan A result gets reported. The hardware must pass incoming inspection for issue, condition and host compatibility, including confirmation that the device enumerates as a USB video-class source on the Raspberry Pi and that its field, focus and exposure can be locked and read back. A new versioned optical calibration must exist, because the pixel scale changes with the sensor and lens, and without it no millimetre value holds. The supervised fault tests behind the G5 safe-output release gate must be repeated, because removing the wireless leg changes the acquisition failure modes: Wi-Fi loss disappears as a fault class, while cable seating, USB bandwidth contention and Pi-side capture stalls arrive as new ones. RSK-13 covers the case where delivery or calibration blocks the migration; the interim configuration then stays the delivered path, and the change is reported as not performed rather than assumed.

Removing the wireless leg narrows the risk register too, not just the responsiveness problem. RSK-01 loses its wide-field sampling component, and RSK-03 loses its Wi-Fi latency and stale-buffer components. Migration alone closes neither risk. Multi-frame confirmation, frame-age monitoring and output inhibition all stay in force, and both entries have to be re-rated against measurements from the new chain rather than against the expectation that it performs better.

6. Current Integration Evidence and Limitations

Current work materially reduces integration risk. It also answers a different question from defect validation, and the two should not be confused. Every record below came off the interim OV3660 configuration, so none of it describes the selected close-range chain; Section 5.8 states which of these results migration invalidates. Table 13 records what happened under the documented conditions and, just as importantly, what cannot follow from it.

Table 13. Current integration evidence, permitted inferences and claim boundaries.

Evidence and label	Exact recorded observation	Allowed inference	Does not establish
Week 4 camera-link record, `LIVE_INTEGRATION_ONLY` (ENG5105 Project Team, 2026h)	At 640 × 480, 792 frames arrived in 30.056 s, giving 26.35 FPS in the hardware integration test.	Physical camera identity, MJPEG	Final SVGA throughput or any rPLA result.

Evidence and label	Exact recorded observation	Allowed inference	Does not establish
		connectivity and source-adapter operation.	
Resolution and frame-rate record, `LIVE_INTEGRATION_ONLY` (ENG5105 Project Team, 2026f)	VGA recent window ≈10.48 FPS; SVGA windows 14.74, 15.60 and 13.96 FPS plus 10.77 FPS over a sustained 30 s test; XGA windows 9.72, 7.51 and 9.42 FPS. The displayed drop field was 0, but the resident counter was then initialised to zero and never incremented. SVGA 800 × 600 was selected.	An operating profile and exposed throughput constraint under the recorded no-filament condition.	Zero frame loss, R4, clarity or defect sensitivity; no raw per-frame rate trace is frozen.
Live-path brightness-step record, `LIVE_INTEGRATION_ONLY` (ENG5105 Project Team, 2026b, 2026d)	Ten reversible transitions gave 101.32 ms median, 222.99 ms P95, 87.12 ms minimum and 222.99 ms maximum. With only $n = 10$, the reported P95 equals the sample maximum and is a weak tail estimate. Brightness returned to 1; service state remained `HEALTHY`.	Camera-control request → OV3660 → Wi-Fi MJPEG → Python processing → local API JPEG response in that small integration sample.	Sensor-exposure-to-browser-paint latency, a stable production P95 or R5.
Live-path network-fault record, `LIVE_INTEGRATION_ONLY` (ENG5105 Project Team, 2026b, 2026c)	Scoped outbound blocking to camera `172.20.10.2` was recognised in 654.64 ms as `RECONNECTING`; detection and the development `automatic_actions_allowed` field were `false`; a 4 s hold added 0 frames; recovery to `HEALTHY` took 5689.35 ms; the firewall rule was removed.	The tested Windows path gated candidate processing and recovered from this injected fault.	Actuator authorisation, final Pi/Wi-Fi reliability, STM32 wiring or certified safe action.
Week 3 contract record, `OFFLINE_SYNTHETIC_ONLY` (ENG5105 Project Team, 2026g)	`FramePacket` → `BaselineVisionProcessor` → `DetectionRecord` → SQLite/JSON/CSV/annotated frames; physical fields are rejected without `calibration_id`.	Data-flow, provenance and null-field safeguards operate on repeatable	Classification of real bubbles, pits, scratches, cracks, contamination or inclusions.

Evidence and label	Exact recorded observation	Allowed inference	Does not establish
		synthetic inputs.	
Week 4 policy-primitive record, `OFFLINE_FAULT_INJECTION_ONLY` (ENG5105 Project Team, 2026h)	In the offline harness, buffer capacity is 1; no frame for 1 s produces `STALE`; no frame for 2 s produces `DISCONNECTED`; three consecutive valid frames are required after reconnection. The resident service now instantiates the same primitives behind a separate acquisition thread.	The one-slot buffer and 1 s/2 s state-policy primitives pass offline fault injection, and their service integration passes threaded unit tests on the development host.	Live backlog suppression under network load, live timeout compliance, full hardware safe-output behaviour or 30 min reliability.

R4 is a capture-to-result target, and the interim numbers have to be read against that rather than against a frame rate. The sustained SVGA observation of 10.77 FPS is a continuous preview rate on the interim chain; it is not a capture-to-result time, and the two are not interchangeable. What the interim figure does bound is how quickly a live preview can settle before an operator triggers a capture. What it says nothing about is the selected chain, where each capture decodes a far larger frame and the relevant quantity is the interval between the capture request and a displayed preliminary result. That interval has not been measured on any configuration.

Timing needs the same restraint. The recorded 222.99 ms P95 sits inside the 500 ms R5 window, but its measurement path has no sensor-exposure timestamp and no browser-paint completion, so it measures a shorter chain than the requirement names. It also rests on ten transitions, where the reported P95 simply equals the maximum. We therefore do not use it as physical alarm-delay evidence, or as a stable estimate of the tail.

7. Calibration and Validation Strategy

Physical rPLA samples and fixture materials are scheduled for delivery on Thursday of Week 5. Consequently, system performance against the intended application target cannot yet be evaluated. Until the materials have been received and their condition verified, all application-performance items will retain the status PENDING_REAL_MATERIAL. The proposed validation methodology follows a sequential evidence-based process: first, the system geometry is fixed: second, the measurement channels are calibrated: third, parameter tuning is conducted exclusively using a separated development dataset: and finally, performance is evaluated once using independent test units. Random allocation of adjacent video frames between the training and test datasets would introduce data leakage because these frames exhibit nearly identical visual characteristics, potentially leading to an overestimation of system performance (Figueiredo & Mendes, 2024; Kapoor & Narayanan, 2023).

7.1 Week 5+ calibration sequence

Upon receipt, each spool and filament segment will be assigned a unique identifier, and the test fixture will be photographed before any detection thresholds and adjust. the camera mount, specimen guides, working distance, focus, exposure, capture profile, region of interest (ROI), background, and bright-field and dark-field illumination settings will subsequently be fixed as a single optical configuration. The capture profile will correspond to the configuration under calibration: 3840× 2160 for the selected imaging chain and SVGA 800 × 600 for the interim chain.

Optical scale calibration will then be conducted. If a traceable reference with length L_{ref} spans P_{ref} pixels in the filament plane, the image scale and candidate-feature length are calculated as

$$s = \frac{L_{ref}}{P_{ref}}\ mm/pixel, \qquad L_{candidate} = sP_{candidate}$$

The optical-scale calibration will be repeated at multiple positions across the ROI and following remounting of the camera or fixture. The procedure will also determine the effective resolution of the installed optical system rather than relying on a theoretical estimate. A resolution target will be imaged at the fixed working position, and the finest distinguishable feature group will be recorded together with the measured pixel scale. This verification is required because a sampling density of 96 px/mm does not, by itself, demonstrate that a 0.20 mm feature can be resolved. Depth of field will be measured under the same configuration by incrementally moving a target through the focal region, as the allowable fixture-placement tolerance depends on this measurement. Only records associated with a valid optical calibration identifier may contain a value in the `size_mm` field. The existing generic `calibration_id` field does not record the illumination mode or rotational index used during image acquisition. Therefore, the database schema must be extended before any physical measurement value is released.

The auxiliary measurement channel will be calibrated after the optical configuration has been fixed because its current response depends on the same system geometry. The five optical-path states defined in Section 5.3 will be measured with both illumination branches de-energised. The channel gain will be selected to prevent saturation, while the low-pass filter corner frequency will be determined from the measured noise characteristics and dynamic response. Raw converter readings will be retained together with the corresponding health-state decision. The illumination reference detector will be calibrated during the same session using a separate reference procedure. With a shite reference target positioned in the field of view, each illumination branch will be adjusted to achieve the required exposure. The corresponding drive level and reference-detector reading will then be recorded as the set point and acceptance band used by the control loop described in Section 5.5. The auxiliary channel will remain limited to material-presence and optical-health monitoring, even if its output shows a correlation with a visible surface anomaly.

7.2 Dataset and event-level evaluation

Ground-truth labels will be assigned to physical sampling units rather than to individual image frames. Both normal specimens and deliberately characterised anomalous specimens will be labelled according to defect class, position within the specimen, measurable physical extent, spool and batch identifiers, optical configuration, illumination mode, rotation index, and the annotator's decision. Before any parameter tuning is performed, complete spools, specimens, or defect events will be allocated to either the development dataset or the independent test dataset. The validation matrix will include challenging but foreseeable variations, such as material colour, surface gloss, illumination mode and intensity, camera remounting, and specimen repositioning. These variations are necessary because real-world anomaly-detection benchmarks indicate that performance measured under a single, narrowly controlled acquisition condition should not be interpreted as

evidence of general system performance (Carvalho et al., 2024; Heckler-Kram et al., 2026; Wang et al., 2024).

Multiple detection boxes associated with the same region of a stationary specimen will be consolidated into a single event based on their spatial overlap within each view. Events observed from different views of the same specimen will subsequently be reconciled across the predefined rotational positions. Using the event-level counts of true positives TP, false positives FP and false negatives FN, system performance will be calculated as

$$Precision = \frac{TP}{TP+FP}, \qquad Recall = \frac{TP}{TP+FN}, \qquad F1 = \frac{2\,Precision\,Recall}{Precision+Recall}.$$

The report pairs these with false alarms per specimen, misses by class and condition, and uncertainty or confidence intervals. Sample counts are set after a pilot and stakeholder review; naming a large sample-size target now would not make the future estimate any more reliable. The minimum independent event count for every accepted class is frozen before the untouched test all the same. We keep every failed case, and we freeze the independent test set before the final run.

These metrics will be reported together with the number of false alarms per specimen, missed events classified by defect type and test condition, and the associated uncertainty or confidence intervals. The required sample size will be determined following a pilot study and stakeholder review, as specifying an unsupported large sample-size target at this stage would not improve the reliability of the subsequent performance estimate. Nevertheless, the minimum number of independent events required for each accepted defect class will be established before the untouched test dataset is evaluated. All unsuccessful detection cases will be retained, and the independent test dataset will be fixed before the final validation run.

7.3 Verification gates and decision rules

Table 14 converts the retained system requirements into configuration-specific verification gates. Each gate defines the required verification method, acceptance criteria, and response to an unsuccessful result.

Table 14. Planned verification gates and acceptance decisions.

Gate	Method and acceptance decision
Optical visibility: R1–R2	Testing will use calibrated reference targets and independently labelled physical defect events under the fixed optical geometry of the configuration being evaluated. The class-specific primary dimensions defined in Section 3.2 will be established before testing. Acceptance requires demonstration of at least three agreed visible-defect classes and the proposed 0.20 mm primary-dimension target under the declared operating conditions. If either criterion is not satisfied, the optical configuration or the relevant requirement must be revised before evaluation using the untouched test dataset.
Classification: R3	Following the pilot study, the minimum number of independent events will be fixed before evaluation. Merged defect events will then be assessed once using the untouched spool-, specimen-, or event-level test partition. For every accepted R1 class satisfying R2, event-level precision and recall must each reach at least 90%. Event counts and confidence intervals will be reported separately for each class. F1 scores and macro- or micro-averaged results must not be used to conceal failure in an individual class.

Gate	Method and acceptance decision
Responsiveness: R4	The durations of illumination soft-start and stabilisation, image capture, decoding, inspection, and result display will be measured separately over repeated single-view acquisitions. Measurement results will include timing jitter, processing stalls, and reconnection events. If the measured capture-to-result duration exceeds 1 s, the capture configuration, ROI, or processing algorithm must be optimised, or R4 must be formally revised with stakeholder approval. The interim preview result of 10.77 FPS cannot be reclassified as evidence of compliance with this requirement.
Alarm path: R5	The event-confirmation rule and number of test trials will be fixed before testing begins. A single synchronised video will record the illuminated inspection region, a millisecond timing reference, and the GUI or physical alarm indication. Alarm latency will be measured from the first frame in which the marked defect is clearly visible, with illumination health already confirmed, to the assertion of the visible alarm. This interval includes event confirmation, processing, and communication delays. Illumination soft-start and settling time will be reported separately under R4 and must not be added to or removed from the R5 interval to alter the result. Sensor-to-GUI and sensor-to-physical-output latency distributions will be calculated separately, and raw video evidence will be retained for each trial. The 95th-percentile latency must not exceed 500 ms for the predefined test set. The existing (n=10) brightness-control-path test does not contribute to R5 acceptance.
Records and endurance: R6–R7	The expected identifiers for all confirmed events will be fixed before the 30 min endurance test. A record will be considered complete only when it contains the corresponding database entry, required evidence image, batch and time fields, specimen and view identifiers, and all applicable optical-calibration provenance. Record loss will be calculated as $(N_{expected} - N_{complete})/N_{expected}$ and must remain below 1%. The dataset will also be audited for orphaned and duplicated records. Source-frame losses and processing-frame losses will be reported separately. Although `dropped_frames` is now measured by the one-slot buffer, this value must not replace event-by-event record reconciliation.
Fault safety and safe-output release: R8	Fault testing will include camera disconnection, blocked or attenuated illumination, forced illumination-mode mismatch, interrupted serial communication, termination of the Raspberry Pi service, and watchdog timeout using the physical wiring. Every injected fault must inhibit the physical output and generate a timestamped fault record. Recovery will require restoration of valid system health and, where the fault is latched, operator acknowledgement. Any unsuccessful fault test will prevent passage through G5 and require the automatic output to remain inhibited.
Privacy and enclosure: R9	Every final-evidence frame that is stored or served will be inspected, together with the complete camera field of view, the enclosure, and the local storage and access configuration. Frames containing a person or image regions outside the enclosed filament tunnel will not be accepted. External publication will remain disabled by default (Australian Government, 2025; Office of the Australian Information Commissioner, 2026). The current full-frame development preview does not satisfy this verification gate.

Acceptance applies only to the specific system configuration under evaluation. Any modification to the camera position, field of view, focus, exposure, illumination geometry, ROI, sample-holder geometry, event-processing logic, or model will invalidate the affected calibration and require regression testing in accordance with the project lifecycle record (ISO/IEC/IEEE, 2023). A physical performance claim will be released only when the associated requirement, optical provenance, independent dataset, stable evidence-record identifier, verified SHA-256 hash, result, and stated limitation are fully traceable. Until these conditions have been satisfied, the system will be classified as an instrumented candidate detector supported by limited integration evidence, rather than as a validated rPLA quality-monitoring system.

8. Environmental, Safety, Ethical and Governance Implications, including Procurement

A recycled feedstock does not make the monitor environmentally beneficial by default. Earlier warning may avoid further processing or use of visibly non-conforming filament, which supports the intent of Sustainable Development Goal 12 (United Nations Department of Economic and Social Affairs, 2026). The camera, Raspberry Pi, lighting and network all consume energy, though; added electronics carry embodied impacts, and a poorly tuned detector can create avoidable rejects. No life-cycle assessment has been completed. Energy and reject observations are reported separately, and we will not claim a net benefit without a defined ISO 14040-compatible comparison (International Organization for Standardization, 2006). Reusing university hardware and choosing repairable fixture parts are design choices, not proof of environmental superiority.

Responsible operation means keeping uncertain measurements out of automatic quality decisions. A stale frame, a missing heartbeat, an unhealthy camera, absent calibration or a lighting mode that does not match the one requested all inhibit rejection and stop requests. The proposed STM32 watchdog provides independent supervision. Fuses, reverse-polarity protection, local decoupling and appropriate flyback suppression address the electrical hazards. The sample holder and the demonstrator output need guarding, low-energy commissioning and an accessible independent stop. These measures are informed by ISO 13849-1 (International Organization for Standardization, 2023), but the prototype is not safety-certified and must not be presented as an industrial safety-related control system.

The camera creates a different obligation. The release configuration views an enclosed filament region, crops or masks locally to the tunnel, and saves event evidence rather than continuous video. The current development endpoints still serve a full annotated frame with an ROI overlay, so they do not satisfy R9. Access, retention and deletion rules have to be agreed before data collection, with laboratory notice and controlled export; OAIC guidance and the Privacy Act give the context (Australian Government, 2025; Office of the Australian Information Commissioner, 2026), though this proposal makes no legal-compliance finding. Any frame containing a person is quarantined from the defect dataset pending review.

Configuration governance is what separates a convincing demonstration from reproducible evidence. We version camera settings, ROI, software, calibration files, requirement status and protocol as one baseline. A reported result needs its evidence label, its conditions and its configuration. `size_mm` stays null until valid optical calibration provenance exists, and spool, batch or event-group partitions are fixed before tuning. An acceptance evidence package must assign a stable evidence-record identifier and store a SHA-256 for raw data, logs, the database, images and video, test manifests and scripts. Consultation outcomes, requirement changes, failed tests and waivers enter a decision log that never assumes stakeholder approval, applying the traceability intent of ISO/IEC/IEEE 15288 (ISO/IEC/IEEE, 2023) and OT security's separation of operational, network and human controls (Stouffer et al., 2023). The current `/api/events/delete` route is a

development-only hard deletion; it now writes an audit event naming the removed identifiers, but it still lacks authentication, a deletion reason and tombstones, so it must be disabled before evidence release. Deletion and export tests belong to the R9 audit. Released evidence will use role-identified quarantine/tombstone records with reason, time and operator identity; corrections will be appended rather than silently substituted, and no existing evidence chain may be erased.

Procurement uses the same discipline. The Plan A camera is approved at A$142.00 through the team's own supply channel and has not been ordered. A live retail page is a weak evidence base for either price or availability: its stock state and shipping options change without notice, and the page keeps no record of what it said on the day it was read. The citation (Banggood, 2026) is therefore used for package contents and sensor identity, while price and availability of record come from the team's channel, and what survives review is a dated screenshot or a formal quotation held in the evidence bank rather than a live URL. The fixture, auxiliary power, illumination optics, calibration and demonstrator groups carry team planning allowances instead of quotations, and Section 9 states them as ranges with that standing attached. Three claims hold up the Plan A scores in Table 6, and each becomes a gate rather than an assumption. Two of them are answerable before money moves. The device must be confirmed to present a standard USB video-class interface to Linux on the Raspberry Pi 4, because the listing names Windows and macOS only and a supplier's assurance is not a test. Its field adjustment must be confirmed to be optical rather than a digital crop, because the Section 5.2 sampling density assumes it. The third cannot honestly be gated before purchase, because it needs the device in hand: the delivered resolving power at the frozen working point, measured against a resolution target at G3, decides whether the arithmetic sampling density of 96 px/mm corresponds to a legible 0.20 mm feature. Failing either pre-order check stops the order. Failing the resolving-power measurement does not stop the build, but it forces a narrower field, a revised R2 or a different optical head, and it must be reported rather than absorbed. The approved purchase also needs a fresh stock, delivered-cost and delivered-lead-time check before the order goes in.

Electrical release uses official component limits rather than the equipment-list shorthand. Raspberry Pi documents its GPIO and alternate-function interfaces as 3.3 V (Raspberry Pi Ltd, 2026b); Microchip specifies the supplied MCP3208's operation, SPI and V_{REF}-bounded analogue input, which is the comparison behind using the microcontroller's own converter instead (Microchip Technology, 2008); and the LM324 data sheet establishes the supply and non-rail-to-rail output behaviour that must be allowed for in the conditioning stage (Texas Instruments, 2025). These sources support the Section 5.3 interface constraints. They do not verify the proposed wiring or component values.

9. Bill of Materials

Table 15 shows the standing of every figure: university-supplied equipment, equipment already held, the team-reported A$40.00 camera price and five planning allowances. All amounts are Australian dollars; the camera's shipping and tax treatment remain to be confirmed. A planning allowance is not a quotation and is replaced by a dated supplier record before commitment. Physical issue, condition and compatibility of supplied parts still require an incoming check (Monash University, 2026c).

Table 15. Bill of materials for the selected build, including the items shared with the interim chain.

Status	Item or group	Qty.	Purpose in the build	Cost basis	Cost charged
University supplied	Raspberry Pi 4 4 GB, power supply and 32 GB SD card	1 set	OpenCV service, API/WebSocket, SQLite and GUI (Raspberry Pi Ltd, 2026a)	Supplied; condition to be checked	A$0
University supplied	STM32 Nucleo-L432KC	1	Heartbeat, capture gating, lighting and inhibited output interface (STMicroelectronics, 2026)	Supplied; condition to be checked	A$0
University supplied	XIAO ESP32S3 Sense with OV3660	1	SVGA MJPEG acquisition node (Espressif Systems, 2026; OmniVision Technologies, 2011; Seeed Studio, 2026)	Supplied; condition to be checked	A$0
University supplied	SFH4555, BPW46 and LM324N, with the MCP3208 held as a spare	1 channel/set	Detect filament presence and check the condition of the optical path using the STM32	Supplied; final component values pending measurement	A$0
University supplied	I2C LCD, LEDs, potentiometers, resistors, breadboard, protoboard, wire and headers	As issued	Local status, conditioning and prototype assembly	Supplied; quantity/condition to be checked	A$0
Existing equipment	LED strip lighting set	1 set	Controlled bright-field and low-angle dark-field illumination	Already purchased and held; no further procurement	A$18.00
Planning allowance	Enclosed 12 V DC supply with screw terminals, 5 V conversion, MOSFET	1 set	Provide separate power for the auxiliary system and safe switching for the connected loads.	Team allowance for an Australian-sourced set; ratings and quotation pending	A$80.00–120.00

Status	Item or group	Qty.	Purpose in the build	Cost basis	Cost charged
	drivers, fuses, connectors and protection				
Planning allowance	Enclosure, camera bracket and indexed sample holder	1 set	Enclosed optical geometry and repeatable sample placement	Team allowance; dimensions pending receipt of the camera and material, and the rotational indexing carries the upper end	A$40.00–90.00
Planning allowance	Diffuser, crossed-polariser set and illumination reference detector for the LED strip	1 set	Even bright-field illumination, specular suppression on a glossy cylinder, and closed-loop feedback on lamp output	Team allowance; film sizes pending the head geometry, and the detector drops out if the issued kit contains a spare photodiode	A$30.00–70.00
Planning allowance	Optical calibration set: graduated scale, resolution target, dark and white references	1 set	Pixel-scale calibration and measured resolving power at the frozen working point	Team allowance; target grade pending the measured field	A$40.00–90.00
Planning allowance	Reject/stop demonstrator	1	Protected physical-output demonstration	Team allowance; purchased with the rest of the build, with its physical output released only after G5 and an authorised validated event rule	A$10.00–25.00
Approved purchase	ALC-IMX258-LY0001-ZB V1 USB 2.0 camera module	1	Selected-concept image acquisition (Supplier product parameter sheet, 2026)	A$40.00 reported by the team on 27 Aug.; quotation, delivery charges and receipt pending	A$40.00

Status	Item or group	Qty.	Purpose in the build	Cost basis	Cost charged
Additional procurement			The camera plus the five allowance groups; compute is supplied and the strip is already held	**One team-reported price and five team allowances**	**A$240.00 –435.00**
Build total			Additional procurement plus the strip already paid for	**Two known/reported figures and five team allowances**	**A$258.00 –453.00**

The camera sits in the same table as everything else because there is one build budget. Its status marks it as the selected concept-specific purchase whose approval and delivered evidence are still pending.

The total is a range because five charged lines are allowances. The camera is recorded at the team-reported A$40.00 and the held LED strip at A$18.00. The five allowances span A$200.00 to A$395.00, placing additional procurement at A$240.00 to A$435.00 and the build total at A$258.00 to A$453.00. A delivered quotation may move those figures, so every allowance and the camera price are restated when dated records exist.

Before commitment, every charged item requires a fresh availability, compatibility, delivered-cost and delivered-lead-time record; the present Plan B evidence does not establish delivery timing.

9.1 Purpose and intended use of each procured item

Table 15 records what each item costs and where that figure came from. This section records what each item is for and how the build will use it, which at this stage matters more than the price. Nothing below is a measured or calibrated value. Every quantity that governs performance — gains, thresholds, pixels per millimetre, working distance, drive currents, fuse ratings — is fixed later under the Section 7 calibration sequence, and none of it is claimed now.

LED strip lighting set. Surface defects on a translucent cylinder are visible or invisible depending almost entirely on how they are lit, and the project briefly names lighting control as a design challenge for this quality parameter. The strip is the controllable source that makes illumination a design variable rather than whatever the room happens to provide. It is mounted inside the enclosed head in two roles: bright-field, aimed to bring out colour and texture contrast, and low-angle dark-field, aimed across the surface so that relief casts shadow. The STM32 switches it through a protected driver, so illumination state is recorded alongside every frame and loss of lighting becomes a detectable fault rather than a silent change in image appearance. Geometry, angle, drive current and which mode suits which defect family are experimental questions for Section 7. The strip exists so that those questions can be asked at all.

The auxiliary power system includes an enclosed 12 V DC supply, 5 V conversion, MOSFET drivers, fuses, connector and protection. It is kept separate from the computing power supply so that switched loads do not affect the Raspberry Pi. The screw-terminal supply is used because several loads need separate and secure connections. The 12 V supply powers the two lighting branches and the demonstrator, while a buck converter provides 5 V for the acquisition node and analogue devices. MOSFET drivers allow the STM32 to control the lighting and demonstrator without connecting these loads directly to the controller pins. Fuse, reverse-polarity protection and

labelled isolation are added for electrical safety. The final ratings, fuse values and wire sizes will be decided after G2 testing.

Enclosure, camera bracket and indexed sample holder. These make the optical measurement repeatable. The enclosure shrouds the head so that ambient light does not enter the frame and so that stored images contain only the sample tunnel, which is what R9 asks for. The bracket holds the camera rigidly, because a working distance that shifts invalidates the optical calibration it was measured at. The holder is where the design earns its depth of field: it grips the specimen outside the imaged region, sets it on the optical axis to a tolerance the measured focus depth can absorb, and indexes it through the six rotations Section 5.2 specifies so that circumferential coverage is a mechanical guarantee rather than an optical hope. Holder geometry, index repeatability and the achievable placement tolerance are all Section 7 measurements.

Diffuser, crossed-polariser set and illumination reference detector. A 1.75 mm strand of filament is a glossy cylinder, and a glossy cylinder lit from one direction returns a saturated highlight down its length with darkness on either side of it. No amount of exposure control recovers surface detail from a saturated band. The diffuser spreads the existing strip into an even bright field so that the highlight becomes a broad gradient rather than a blown line; the polariser pair, one film over the source and one over the lens with their axes crossed, suppresses what remains of the specular return and leaves the diffuse component that actually carries texture and colour. Both are cheap film-and-frame items rather than instruments, and both change the illumination geometry, so either one entering or leaving the head invalidates the optical calibration recorded with it. Which combination suits which defect family is a Section 7 question, and the low-angle dark-field mode is deliberately left unpolarised, because relief is what it is meant to reveal. The reference detector belongs in this group rather than with the auxiliary channel, because what it has to watch is this assembly: a second photodiode looking at the diffuser output, on its own conditioning stage and its own converter channel, giving the Section 5.5 loop a reading that follows lamp output instead of following the specimen. It costs a few dollars and it is what stops one detector from being asked whether the light changed or the sample did.

Optical calibration set. This is the smallest line in the build and the one the largest claims rest on. A graduated scale imaged at the frozen working point fixes millimetres per pixel, without which `size_mm` stays null and R2 has no meaning. A resolution target imaged at the same point fixes something the scale cannot: whether the optics actually resolve what the pixel count implies. Section 5.2 puts the sampling density at 96 px/mm and states plainly that this is not a resolving power, and the target is how that gap gets closed with a measurement rather than an assumption. The dark and white references sit in the same holder so they can be inserted without moving the camera, and they reveal saturation, drift and uneven illumination that a picture of filament alone would hide.

Reject/stop demonstrator. This is the project's physical output, and it is what separates a detector from a monitoring system that can act. When an event is confirmed and every authorisation condition holds, the STM32 drives the demonstrator to signal the reject or stop decision the confirmed event calls for. It is also the test object for the whole safety argument: fault injection under R8 means something precisely because a real actuator exists that must stay inhibited when heartbeat, calibration or lighting health is lost. Its physical output stays disabled until the G5 release gate passes.

ALC-IMX258 USB 2.0 camera module. This replaces the interim acquisition front end. The supplier lists several lens fields and AF/FF variants, but the build must select one, provide its own rigid bracket and controlled lighting, and prove that focus and exposure stay fixed during measurement (Supplier product parameter sheet, 2026). The module reaches the Pi over a local USB cable, removing the interim wireless leg. Its rolling shutter is managed by holding the

specimen still and enforcing a settling interval. Section 5.8 records every item that must be re-established after the swap; no interim timing or image-quality result carries across.

10. Project Delivery Plan and Team Responsibilities

The delivery plan follows a stage-gate process within a V-model structure. User needs are translated into proposed requirements, interfaces, hazards and test definitions; implementation then proceeds through unit checks, subsystem integration, calibrated system testing and independent validation against the corresponding requirements IDs. A gate is not passed simply because the schedule has moved forward. It is passed only when the evidence specified for the stage has been produced, reviewed and assigned to its owner.

Two separate dates control the sequence and neither depends on the other. Plan A remains the plan of record so the selected camera is built into S1-S6 rather than left until the end. Before calibration starts, gate GP determines whether the main sequence will run on the camera. Plan B is not a parallel build competing for the same project weeks. Instead, it carries the software interface and safety de-risking work that both chains require, and becomes the acquisition front end for S2-S6 only if GP fails. This keeps the fallback inexpensive because everything above the frame contract has already been built and validated so a failed GP consumes optical-calibration time rather than forcing a redesign. The second controlling date concerns material availability; the evidence status changes on Thursday 27 August 2026 (Week 5). Before that date, the project cannot perform physical rPLA or fixture-material testing. Earlier activities are therefore restricted to design, procurement preparation, software development, synthetic testing, and bounded live integration that does not use rPLA. If the material arrives late, incomplete, damaged or unsafe, the physical milestones must be delayed accordingly, because synthetic results cannot substitute for the missing physical evidence. Table 16 makes this dependency explicit. It functions as a release gate rather than as a hidden scheduling assumption.

Table 16. Stage-gated delivery schedule and evidence decisions.

Stage and timing	Planned work	Gate output and decision rule
S0: baseline, through Wed. 26 Aug.	Freeze requirements, interfaces, risks, fixture drawings and protocols; retain synthetic/live-integration labels	G0: owner-reviewed versioned baseline; no physical-performance release
GP: procurement decision gate, order by Wed. 26 Aug., arrival by Mon. 31 Aug.	Record approval, exact lens/focus option, A$40.00 price basis, delivered cost and arrival plan; then run incoming Raspberry Pi and optical-control checks	**GP decides which chain S2–S5 run on.** A missed arrival, failed Pi enumeration/power test, unstable control lock or unsuitable measured field keeps the interim chain in S2–S6 and moves Plan A to stage SA
G1: material gate, Thu. 27 Aug.	Record receipt, quantity, condition, dimensions, traceability and hazards	Pass releases physical work; failure reschedules it and reviews scope
S1: safe shakedown, after G1	Assemble fixture; check guarding, power, communications and inhibited outputs at low energy	G2: completed safety checklist, stable acquisition, no uncontrolled output

Stage and timing	Planned work	Gate output and decision rule
S2: calibration, Weeks 6–7	Lock optics on the chain GP selected; establish pixel scale, delivered resolution, depth of field and references	G3: versioned IDs; invalid calibration keeps millimetre fields null. The calibration names the chain it was taken on and does not transfer to the other
S3: acquisition, Weeks 7–8	Collect labelled events across the planned illumination modes, rotations and optical challenges; audit labels and partitions	G4: traceable ground truth and independent groups before tuning freeze
S4: robustness, Weeks 8–9	Integrate subsystems; inject supervised network, power and sensor faults	**G5 safe-output release gate:** every R8 case inhibits physical output and records a timestamped fault; any failure or unresolved high risk keeps output inhibited
S5: validation, Week 10, after G5	Evaluate event metrics, false alarms per specimen, capture-to-result time, latency and record loss on held-out groups, on the chain GP selected	Accept, optimise, narrow the envelope or revise a requirement; report failures. Automatic output still requires an authorised validated event rule and all live health/calibration gates
S6: release, Weeks 11–12	Freeze configuration, assemble evidence and peer-review claims	Register links each accepted claim to its requirement ID, configuration, stable evidence-record identifier and SHA-256, test and owner
SA: migration, only if GP failed, on later delivery	Inspect the camera on arrival and confirm that it enumerates as a stable Linux video device on the Raspberry Pi and exposes the supplier-claimed MJPEG/YUY2 capture modes; fit the USB front end behind the existing capture contract, re-calibrate, and repeat the S2–S5 protocol on the selected optical chain	GA: migration accepted only after incoming inspection, a new versioned calibration and repeated G5 fault tests. No Plan A result may be reported before its own calibration exists, and interim figures are never carried across

This sequence puts the responsiveness gap on the table. The recorded SVGA integration test observed 10.77 FPS under the `LIVE_INTEGRATION_ONLY` evidence state (ENG5105 Project Team, 2026f), which is a preview rate on the interim chain rather than a capture-to-result time. The independent-validation gate therefore demands one of three things: optimisation, a narrower field and capture envelope, or a documented requirement change. A retrospective claim of compliance is not on that list. Stage SA measures capture-to-result on the selected optical chain after migration, since the USB link takes the Wi-Fi leg out of the path while 4K decoding adds Pi-side cost the interim profile never carried. We assume no improvement in advance. Any figure for the selected chain comes from measurements taken after its own calibration, and the interim result never gets restated as a Plan A result.

Table 17 maps each role identifier to the member who holds it; the identifiers are reused elsewhere in this report to keep the ownership columns compact. The planned workload shares in Table 17 sum to 100% and exist to guide balancing. They do not determine individual marks.

Table 17. Proposed team responsibilities, workload shares and contribution evidence.

Portrait	Member, role and share	Primary accountability	Fixed support, review and required evidence
	**M1 — Letu Ma** — systems/project; 17%	Requirements, stakeholder protocol, architecture, schedule, interfaces, risk and integration	**Support/review:** M4 on alarms and acceptance; all module owners. **Evidence:** Decision log, baseline and requirement-status records.
	**M2 — Tingyu Zheng** — optics/camera; 16%	Camera, lighting, enclosure, coverage, image quality and calibration	**Support/review:** M4 on images and ROI; M6 on optical repeats. **Evidence:** Drawings, settings, calibration and image records.
	**M3 — Zhiqian Wang** — sensing/power; 16%	Photodiode/ADC, power, protection and electrical inspection	**Support/review:** M5 on wiring and faults; M4 on health logs. **Evidence:** Schematics, calculations, inspections and logs.
	**M4 — Xinyang Wu** — Group leader, vision/UI/data; 18%	OpenCV, GUI, database, tracking, evaluation and analysis	**Support/review:** M2 on imaging; M6 on independent labels and tests. **Evidence:** Code, schema, test output and reproducible analysis.
	**M5 — Niewei Fan** — embedded/comms; 17%	STM32, capture gating, lighting modes, timing, watchdog, communications and safe output	**Support/review:** M3 on power; M6 on mechanical and output tests. **Evidence:** Firmware, interface traces and fault records.
	**M6 — Haowei Ding** — validation/mechanical/quality; 16%	Rig, ground truth, group split, validation, uncertainty and safety review	**Support/review:** M4 on metrics; M5 on alignment and repeatability. **Evidence:** Protocols, label audit, held-out results and review.

Integration stays shared. Every work item names an owner, a reviewer, an internal date and an evidence output. Any interface change takes a version increment, an owner review and a recorded impact check before it joins the baseline. Released calibration and test data are read-only: corrections get appended rather than substituted, and the development hard-delete route is excluded from the release configuration. We will run a weekly integration build and raise the cadence from G3 through the independent-validation gate, which keeps hand-offs visible without letting M1 or M4 become the default owner of whatever is unfinished.

11. Risk Register

A risk register is only valuable when the team can recognize a problem before it occurs. Table 18 therefore links each proposal-stage risk to a visible trigger, an assigned owner, a control measure and a rating. The ratings follow the discipline brief's matrix, combining a probability category–Certain, Likely, Possible, Unlikely, Rare or Eliminated–with a harm-severity category–Negligible, Marginal, Critical or Catastrophic–to produce an overall rating of Low, Medium, High or Very High. Each row shows both the probability-severity pair and the resulting rating so that the assessment can be verified rather than simply accepted. The probability values are qualitative engineering judgments, not measured occurrence rates. No risk is classified as Eliminated because none of the hazards has yet been physically removed. Residual ratings assume that the stated controls have been implemented and tested; they do not indicate that the underlying Hazard no longer exists.

Table 18. Proposal-stage risk register rated on the probability × harm-severity matrix.

ID	Risk and consequence	Initial probability × severity → rating	Observable trigger	Owner	Planned control and contingency	Residual probability × severity → rating
RSK-01	Compression, glare or specular highlights resemble or hide defects	Likely × Critical → High	Candidate rate changes sharply with JPEG or illumination setting; edge shape shifts between views of the same specimen	M2/M4	Lock exposure and geometry, test both illumination modes, confirm across indexed rotations, retain raw event evidence; narrow the operating envelope if needed	Unlikely × Critical → Medium
RSK-02	Glare or incomplete angular coverage causes false alarms or missed surfaces	Likely × Critical → High	Saturated ROI, unstable background histogram or repeatable blind sector	M2	Enclosed bright-/dark-field lighting, matte background, mirrors only after measured coverage test; report uncovered angles	Unlikely × Critical → Medium
RSK-03	Wi-Fi loss, latency or stale buffering corrupts online decisions (interim chain;	Likely × Critical → High	Frame age exceeds threshold, health endpoint fails or backlog grows	M5/M4	The one-slot buffer, age/state monitor and drop instrumentation now sit in the resident service and pass threaded unit tests; **still proposed:** Pi	Unlikely × Critical → Medium

ID	Risk and consequence	Initial probability × severity → rating	Observable trigger	Owner	Planned control and contingency	Residual probability × severity → rating
	narrowed by migration)				deployment, a controlled 2.4 GHz network and STM32-held output inhibition on real wiring. No live guarantee is claimed yet	
RSK-04	Week 5 materials are late, incomplete or unsuitable	Possible × Catastrophic → Very high	Receipt gate fails on 27 Aug. or fixture dimensions conflict	M1/M6	Inspect at G1, keep pre-gate work non-physical, re-sequence software/documentation, escalate scope and dates without fabricating tests	Possible × Critical → High
RSK-05	Too few representative defects produce weak or biased validation	Likely × Catastrophic → Very high	Required classes/groups are absent or labels cannot be independently checked	M6/M4	Predefine sample/label protocol, retain interpretable OpenCV baseline, report class coverage and narrow claims; seek further material through approved channels	Possible × Critical → High
RSK-06	Adjacent-frame or event leakage inflates performance	Possible × Critical → High	Frames from one spool, batch or event appear in tuning and test sets	M6	Freeze group-level partitions before tuning, store provenance, run a partition audit and reject contaminated evaluation	Rare × Critical → Medium
RSK-07	Calibration drift makes size values invalid	Likely × Critical → High	Focus, working distance, field setting or configuration	M2/M5	Bind provenance-verified optical calibration IDs to records, perform reference checks,	Unlikely × Critical → Medium

ID	Risk and consequence	Initial probability × severity → rating	Observable trigger	Owner	Planned control and contingency	Residual probability × severity → rating
			hash changes; reference check fails		invalidate physical fields and recalibrate after change	
RSK-08	Power noise, undervoltage, inductive switching or a 5 V analogue output resets, damages or drives unsafe output from the controllers	Possible × Catastrophic → Very high	Pi undervoltage/reset log, rail sag, conditioned analogue input outside the microcontroller's range, MCU reset or output pulse during switching	M3/M5	Separate power domains, clamp and limit the LM324 output to the STM32 analogue input range, fuses, decoupling, flyback suppression, measured inrush/steady-state/fault tests and watchdog-safe defaults	Unlikely × Catastrophic → High
RSK-09	The sample holder or demonstrator creates pinch, entanglement or unexpected-motion hazards	Possible × Catastrophic → Very high	Guard open, alignment jam, watchdog loss or output occurs outside supervised command	M6/M5	Guarding, low-energy commissioning, independent stop, supervised tests and inhibit output until G5 plus an authorised validated event rule and all live gates pass	Unlikely × Catastrophic → High
RSK-10	Camera captures people, or evidence is retained, accessed or hard-deleted without an agreed rule	Possible × Critical → High	Face/body pixels outside the ROI appear in stored/served output, unapproved export, the development delete endpoint remains enabled, or no	M1/M4	Enclose and crop the view, local event-only storage, notice, role-based access; disable the unaudited hard-delete route and quarantine incidental images	Rare × Critical → Medium

ID	Risk and consequence	Initial probability × severity → rating	Observable trigger	Owner	Planned control and contingency	Residual probability × severity → rating
			retention/deletion decision has been made			
RSK-11	Interface or configuration mismatch prevents late integration	Likely × Critical → High	Schema/baud/state mismatch, unversioned setting or module passes alone but fails baseline build	M1 with all owners	Interface-control record, contract tests, versioned configurations, weekly integration and owner-reviewer sign-off	Unlikely × Critical → Medium
RSK-12	Camera or common-part procurement arrives late or proves incompatible with the Raspberry Pi, delaying the fixture and validation	Possible × Catastrophic → Very high	No approved delivered quote/lead time by the procurement checkpoint, stock changes, or incoming electrical/mechanical checks fail	M1/M3/M6	Obtain fresh stock, compatibility, delivered-cost and delivered-lead-time evidence; identify approved substitutes before order; repeat interface/power checks after any substitution and re-sequence without fabricating physical tests	Possible × Critical → High
RSK-13	The selected camera arrives too late to migrate, fails to enumerate on the Raspberry Pi, or cannot be calibrated on the fixture, so the project delivers on the interim chain	Possible × Marginal → Medium	No confirmed dispatch by the procurement checkpoint, no USB video-class device node on the Pi, or field and working distance cannot be set on the fixture	M1/M2	Keep the interim configuration fully validated so no requirement depends on migration; confirm host compatibility and field adjustment before ordering; report the migration as not performed rather than presenting interim figures as Plan A results	Rare × Marginal → Low

The two ratings change for different reasons, and this distinction reflects the mitigation approach. The severity of physical harm remains unchanged. RSK-08 and RSK-09 therefore remain Catastrophic because guarding, an independent stop, and watchdog-safe defaults reduce the likelihood of entering a hazardous state rather than changing the consequences once that state occurs. Their residual High rating is precisely why actuation remains inhibited until the G5 release gate instead of being allowed simply because the rating has improved. Schedule and material risks are treated differently, with mitigation reducing severity rather than probability. RSK-04, RSK-05 and RSK-12 are reduced from Catastrophic to Critical because re-sequencing and narrowing the claims allow a defensible deliverable to remain possible when a part or material does not arrive, even though the arrival date itself remains outside the team's control. RSK-06 and RSK-10 are the only two cases in which a procedural control genuinely reduces the event likelihood to Rare, because frozen group partitions and an enclosed, release-cropped view remove the mechanism itself rather than merely reducing its consequences.

The stop rules are especially important. RSK-04 prevents physical testing from proceeding. RSK-06 invalidates the affected performance estimate. RSK-07 requires physical-value fields to be reset to null. RSK-08 or RSK-09 prevents actuation. RSK-12 may shift the relevant gate and validation schedule, while RSK-13 may cancel migration without changing any interim acceptance decision. A residual High rating for data or schedule risk is accepted only by narrowing the associated claim or revising the plan through the decision log, never by reducing the evidence label.

12. Recommendation and Conclusion

Plan A remains the recommended concept. It uses the ALC-IMX258 USB 2.0 module to image a held, manually indexed filament specimen, while the Raspberry Pi detects defect candidates and estimates their calibrated size and approximate datum-relative location. The STM32 supervises lighting, watchdog functions and safe I/O. Although the supplier-listed lens and focus options indicate useful close-range imaging potential, this remains conditional on the exact delivered configuration and subsequent Raspberry Pi testing(Supplier product parameter sheet, 2026). Plan B, which uses the supplied ESP32S3 OV3660 to transmit controlled 2.4 GHz MJPEG video, remains the interim implementation and preserves downstream software and data contracts while Plan A procurement and incoming inspection gates are completed. Approval, availability, shipping and lead-time evidence remain outstanding; therefore, these estimates support project sequencing but do not authorise procurement.

The recommendation supports a staged development process rather than immediate system acceptance. Plan B May proceed only after the 27th of August material gate has been passed, followed by safe-assembly inspection, version-controlled optical calibration, traceable acquisition using real material, held-out validation, supervised fault testing, and the G5 safe-output release gate. Precision, recall, F1 score minimum detectable physical defect size, delivered optical resolution, and final processing latency have not yet been verified, while stakeholder approval of numerical requirements still requires documented consultationIf any gate is not passed, the appropriate response is to optimize the system, narrow its operating envelope, or revise the affected requirement with a documented rationale. Plan A is subject to the same evidence requirements, and no performance claim should be made until the delivered Hardware has completed incoming inspection, controlled calibration, and physical validation. Accordingly, this report presents no plan A measurement results. The proposed system should therefore be regarded as a controlled measurement chain with clearly defined State boundaries: image acquisition generates a candidate; valid optical and coordinate calibration may provide an estimated size and approximate fixture-relative position; independent validation may support classification of a defect; and an output action may be requested only when the system is healthy, current, and authorized. Selecting a concept by itself does not demonstrate system performance.

13. Contribution

Xinyang Wu, as the team leader, proposed the overall framework of the project and the structure and main content of the report. Niewei Fan provided constructive suggestions for the plan and proposed optimization suggestions. The other team members revised and completed their respective sections based on the basic content of the plan.

14. AI Use Disclosure

OpenAI's ChatGPT and Anthropic's Claude were used during the preparation of this report to improve grammar and clarity, assist with the construction and formatting of tables, and support the identification and verification of selected important technical information. All AI-assisted content was reviewed and revised by the authors, who remain responsible for the report's technical decisions, interpretation, accuracy, and final content.

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